Data Analytics Capstone

Empirical Modeling of Electric Power Generating Unit Attributes Under the U.S. EPA Clean Air Markets Program via Inferential and Machine Learning Frameworks

Final Report

Dhananjay Verma (Student ID: 0291493)

Walsh College
Department of Quantitative Methods and Data Science

QM640 V1: Data Analytics Capstone
Faculty Advisor: Dr. Javad Katibai
Summer 2026 Academic Term
October 15, 2026

GitHub Repository, Data Assets, and Autonomous Manuscript Notice

To ensure complete computational transparency and empirical reproducibility, the raw federal regulatory dataset, engineered analytical feature matrices, executable Python scripts, JSON validation artifacts, and high-resolution graphical outputs associated with this research are hosted in an open-access GitHub repository:

Primary GitHub Repository URL: https://github.com/djmsritai-tech/QM640-Data-Analytics-Capstone

Source Dataset URI (facility-2025.csv): https://github.com/djmsritai-tech/QM640-Data-Analytics-Capstone/blob/main/data/raw/facility-2025.csv

Engineered Dataset URI (facility_clean_processed.csv): https://github.com/djmsritai-tech/QM640-Data-Analytics-Capstone/blob/main/data/processed/facility_clean_processed.csv

Source Code Directory: https://github.com/djmsritai-tech/QM640-Data-Analytics-Capstone/tree/main/scripts

Visual Figure Archive (300 DPI): https://github.com/djmsritai-tech/QM640-Data-Analytics-Capstone/tree/main/outputs/figures

Project Repository Hierarchy:

```
QM640-Data-Analytics-Capstone/
│
├── data/
│   ├── raw/
│   │   └── facility-2025.csv            [Unmodified EPA CAMPD file: 4,050 units, 30 fields]
│   └── processed/
│       └── facility_clean_processed.csv  [Imputed & transformed matrix: 4,050 units, 45 fields]
│
├── scripts/
│   ├── run_capstone_analytics.py        [Core statistical, econometric & ML execution engine]
│   ├── dhananjay_final_interim_report.py [Automated Interim Report document compiler]
│   └── dhananjay_final_capston_report.py [Automated Final Report document compiler]
│
├── outputs/
│   ├── analytics_results.json           [Serialized test metrics and model parameters]
│   └── figures/                         [Eleven 300-DPI diagnostic figures]
│
└── reports/
    ├── QM 640_Interim_Report_DhananjayVerma-0291493.docx & .pdf
    └── QM 640_Final_Report_DhananjayVerma-0291493.docx & .pdf
```

Autonomous Document Declaration: *This Final Report is authored as a wholly self-contained scholarly treatise. Readers do not require access to prior project milestones to evaluate its methodology or conclusions. Every variable specification, mathematical equation, statistical power proof, architectural diagram, empirical finding, and software routine (Appendices A–D) is documented exhaustively within this volume.*

Abstract

Grid-connected thermal electricity generators monitored under the United States Environmental Protection Agency (EPA) Clean Air Markets Program (CAMP) display substantial structural variation across burner heat input capacity, generator megawatt ratings, commercial age, and post-combustion emission controls. Conventional environmental oversight frequently depends on single-variable heuristics that overlook complex interactions linking thermodynamic cycle design, regulatory grandfathering, and scrubber retrofits. To resolve this analytical limitation, the present study constructs a four-stage quantitative modeling pipeline examining the entire 2025 EPA Clean Air Markets Program Data (CAMPD) facility registry, which encompasses N = 4,050 generating units situated across 1,328 power plants in 52 U.S. jurisdictions. Complete population retention (zero discarded rows) was achieved via regular-expression capacity extraction and technology-stratified median imputation. Four complementary analytical paradigms were deployed: (1) parametric One-Way Analysis of Variance (ANOVA) paired with non-parametric Kruskal-Wallis testing to quantify fuel-based thermal divergence; (2) supervised binary classification comparing Ridge Logistic Regression, Random Forest (150 estimators), and Gradient Boosting (100 estimators) across a stratified 80% training partition (N = 3,240 with 5-fold cross-validation) and a 20% holdout test partition (N = 810) to flag upper-quartile high-emission units (Max Hourly HI Rate >= 2,350.0 mmBtu/hr); (3) bivariate and multivariate Ordinary Least Squares (OLS) econometric models isolating unit vintage effects from physical scale; and (4) unsupervised K-Means partitioning integrated with two-dimensional Principal Component Analysis (PCA). Empirical findings demonstrate that fuel class accounts for 34.91% of variance in thermal input ($F = 542.36$, $p < 0.0001$; $H = 783.02$, $p < 0.0001$), where coal boilers average 5,221.2 mmBtu/hr compared with 1,553.0 mmBtu/hr for natural gas turbines. On out-of-sample holdout testing, Gradient Boosting attained 96.54% accuracy, 94.44% precision, F1 = 0.9303, and ROC-AUC = 0.9865, whereas Random Forest delivered 93.14% recall (capturing 190 of 204 high emitters, ROC-AUC = 0.9851), markedly surpassing linear classification (89.14% accuracy, F1 = 0.8053). Multivariate OLS captured 76.68% of thermal variation ($F = 4,435.0$, $p < 0.0001$), while K-Means uncovered k = 4 distinct fleet archetypes (Silhouette = 0.4431, PCA variance = 80.19%), pinpointing 414 legacy coal baseload units (86.0% high emitters) for prioritized regulatory audits.

Problem: Fossil-fuel electricity generating units exhibit multidimensional nonlinear relationships among boiler technology, fuel chemistry, vintage, and abatement hardware that simplistic univariate screening rules cannot accurately classify.

Solution Approach: A unified four-pillar quantitative architecture integrating parametric/rank variance testing (ANOVA, Kruskal-Wallis, Tukey HSD), ensemble supervised learning (Gradient Boosting, Random Forest, L2 Logistic Regression), multivariate OLS econometrics, and PCA-projected K-Means clustering.

Data: 2025 U.S. EPA Clean Air Markets Program Data (CAMPD) bulk facility registry (facility-2025.csv) covering N = 4,050 generating units across 1,328 plants (30 raw fields expanded into 45 analytical attributes with 0% row loss).

Technology: Python 3.11 scientific computing ecosystem utilizing Pandas, NumPy, SciPy, Statsmodels, Scikit-Learn, Matplotlib, Seaborn, and Python-Docx.

Major Results: Gradient Boosting achieved 96.54% holdout accuracy, 94.44% precision, F1 = 0.9303, and ROC-AUC = 0.9865; Random Forest achieved 93.14% sensitivity (ROC-AUC = 0.9851); Multivariate OLS yielded $R^2 = 0.7668$; K-Means (k = 4) explained 80.19% 2D PCA variance.

Implementation Area: Automated compliance audit prioritization across EPA Regions 1–10, Continuous Emission Monitoring System (CEMS) verification scheduling, and utility coal-to-clean generation retirement planning.

1. Introduction

Background and Context

Airborne emissions generated by utility-scale fossil-fuel power plants represent a central focus of environmental regulation, public health epidemiology, and electricity market design across the United States. Stationary steam boilers and combustion turbines burning coal, natural gas, distillate oil, and biomass supply the dispatchable capacity needed to stabilize the bulk power grid, yet their high-temperature oxidation processes release substantial quantities of sulfur dioxide (SO_2), nitrogen oxides (NO_x), particulate matter ($PM_{2.5}$ and PM_{10}), mercury (Hg), and carbon dioxide (CO_2). Once emitted into the boundary layer, SO_2 and NO_x undergo photochemical and aqueous transformation into secondary particulate sulfates and nitrates, contributing to regional smog, watershed acidification, and cardiopulmonary hospitalization (Harrington et al., 2004; Burtraw et al., 2012).

To curb these negative externalities while preserving grid reliability, federal lawmakers and the U.S. Environmental Protection Agency (EPA) established a series of allowance-trading and performance-standard regimes under the Clean Air Act Amendments of 1990. Prominent among these initiatives are the Title IV Acid Rain Program (ARP), the Cross-State Air Pollution Rule (CSAPR), and the Mercury and Air Toxics Standards (MATS). Administration of these statutes relies on the EPA Clean Air Markets Program (CAMP), which compels regulated generating units to operate certified Continuous Emission Monitoring Systems (CEMS) under 40 CFR Part 75 (U.S. EPA, 2024). The resulting CAMPD repository tracks detailed engineering specifications for every monitored unit, including burner design, generator megawatt capacity, maximum hourly heat input rate, commercial startup date, fuel category, and installed pollution control equipment.

The empirical context of this investigation is the United States utility generating fleet operating in 2025. Economic competition from shale gas, declining wind and solar costs, and stringent air quality rules have reshaped the national fleet over the last twenty years (Schmalensee & Stavins, 2017; Ellis et al., 2019). Consequently, today's grid comprises two contrasting operational tiers: a shrinking yet thermally massive cohort of pre-1985 coal-fired steam boilers equipped with complex post-combustion scrubbers, and a vast fleet of modern natural gas combined-cycle (NGCC) units and simple-cycle peaking turbines built after 2000. Surviving solid-fuel plants routinely combine wet or dry flue gas desulfurization (FGD), selective catalytic reduction (SCR), electrostatic precipitators (ESP) or baghouses, and activated carbon injection (ACI), producing strong structural couplings among unit age, megawatt scale, and abatement density.

Primary stakeholders operating within this domain include EPA regional enforcement officials (Regions 1–10), state air quality control boards, Independent System Operators (ISOs), utility asset managers, and energy transition researchers. These decision-makers require rigorous quantitative models capable of explaining thermal load dispersion, accurately flagging high-emission units for field inspection, and partitioning the 4,050-unit fleet into clear operational tiers to optimize audit budgets and guide orderly plant retirements.

Problem Statement (Single, Measurable; Include Y and X)

Formal Measurable Problem Statement: This study seeks to model, predict, and cluster Y = maximum hourly thermal heat input capacity (Max Hourly HI Rate, expressed continuously in mmBtu/hr) alongside binary upper-quartile high-emission risk status (High_Emitter $\in \{0, 1\}$, triggered when Heat Input Rate

>= 2,350.0 mmBtu/hr) across N = 4,050 regulated power generating units located at 1,328 U.S. facilities, utilizing X = primary fuel classification, boiler/turbine technology, nameplate electrical capacity (Capacity_MW), operational vintage (Unit_Age), cumulative pollution control count (Control_Count), and EPA regional jurisdiction during the 2025 reporting cycle. Analytical performance is assessed via One-Way ANOVA F-ratios, Eta-squared (η^2), and Kruskal-Wallis H-statistics for fuel divergence; out-of-sample Accuracy, Precision, Recall, F1-Score, and ROC-AUC on an 810-unit holdout set for supervised classification; R^2, Adjusted R^2, RMSE, and MAE for econometric regression; and Within-Cluster Sum of Squares (WCSS), Silhouette scores, and 2D PCA explained variance (%) for fleet segmentation.

Although the EPA publishes granular unit-level registries, regulatory compliance screening often relies on one-dimensional cutoffs that ignore how boiler thermodynamics, regulatory grandfathering, and scrubber retrofits interact. Without validated machine learning classifiers and multivariate cluster profiles, inspection resources can be misallocated toward low-risk peaking turbines while overlooking high-intensity intermediate steam units. Bridging this gap demands an integrated methodology uniting classical statistical inference, econometric regression, ensemble classification, and unsupervised clustering.

Purpose of the Study

The objective of this capstone project is to construct and validate a four-tier quantitative analytics framework across all N = 4,050 generating units in the 2025 EPA CAMPD dataset. Specifically, the research is designed to: (1) quantify thermal heat input differences across primary fuel classes via parametric and non-parametric omnibus and post-hoc tests; (2) benchmark regularized linear and tree-ensemble classifiers (Logistic Regression, Random Forest, Gradient Boosting) to predict top-quartile high-emission units and isolate key feature importances; (3) estimate bivariate and multivariate OLS regressions that separate unit vintage effects from physical megawatt scale and abatement density; and (4) partition the national fleet into interpretable operational archetypes using K-Means clustering and PCA.

Research Problems / Research Questions

Four formal research questions (RQ1–RQ4) guide the empirical investigation across the 4,050 generating units:

Research Question 1 (RQ1 — Fuel Variance Inference): To what extent do primary fuel categories (Coal, Natural Gas, Oil/Diesel, Biomass, Other) and combustion technologies drive statistically significant variation in maximum hourly heat input rates (mmBtu/hr), and what fraction of total fleet variance is explained by fuel type?

Research Question 2 (RQ2 — Supervised Emission Risk Classification): How effectively can supervised machine learning classifiers trained on unit-level engineering and regulatory features (Capacity_MW, Unit_Age, Control_Count, fuel class, boiler technology, and EPA region) identify top-quartile high-emission units (Heat Input >= 2,350.0 mmBtu/hr) on unseen holdout data, and which attributes dominate predictive splitting?

Research Question 3 (RQ3 — Econometric Vintage & Scale Analysis): How does generating unit age (years since commercial startup) relate to maximum hourly thermal heat input in unadjusted bivariate models versus multivariate econometric regressions that control for electrical nameplate capacity and installed pollution controls?

Research Question 4 (RQ4 — Unsupervised Fleet Taxonomy): Can unsupervised K-Means clustering and Principal Component Analysis (PCA) partition the 4,050-unit national fleet into cohesive, well-separated operational archetypes with distinct environmental compliance profiles?

Contributions and Expected Value

Practical Contribution: Practically, this work provides an automated risk-scoring pipeline that allows regulatory agencies and utility engineers to replace manual spreadsheet reviews with probabilistic audit targeting, immediately pinpointing aging high-capacity units that warrant intensive CEMS verification or retirement evaluation.

Technical Contribution (Methods/Models): Technically, the study illustrates how combining parametric/rank ANOVA, stratified tree-ensemble classification (Random Forest and Gradient Boosting), multivariate OLS econometrics (resolving omitted-scale confounding), and log-standardized K-Means/PCA clustering within a zero-data-loss Python workflow yields comprehensive analytical insight.

Value to Stakeholders and Target Users: For EPA regional inspectors, the ensemble classifiers deliver 94.44% precision (Gradient Boosting) and 93.14% recall (Random Forest), minimizing wasted field inspections while capturing over 93% of true high emitters. For utility resource planners, the k = 4 cluster taxonomy establishes empirical benchmarks for fleet modernization.

2. Literature Review

Literature Review Approach

To anchor our hypotheses, variable definitions, and modeling choices in peer-reviewed theory, a structured bibliographic search was executed across major scientific repositories—including IEEE Xplore, Elsevier ScienceDirect, JSTOR, SpringerLink, the Journal of the American Statistical Association archive, and Google Scholar—supplemented by federal regulatory manuals from the U.S. EPA. Query logic paired domain descriptors with quantitative algorithm terms: ('Clean Air Markets' OR 'Acid Rain Program' OR '40 CFR Part 75' OR 'power plant emissions') combined with ('thermal heat input' OR 'unit vintage' OR 'scrubber retrofits') and ('analysis of variance' OR 'rank-sum tests' OR 'tree ensembles' OR 'OLS econometrics' OR 'unsupervised clustering').

Selection rules required candidate publications to represent either: (1) empirical energy-policy investigations analyzing unit-level power plant compliance under emissions trading; (2) seminal statistical and machine learning works proving the mathematical properties of ANOVA, Kruskal-Wallis, Random Forests, Gradient Boosting, OLS, and K-Means; or (3) official federal CEMS reporting specifications. Works limited to automotive transport emissions or aggregate global climate projections lacking plant-level resolution were omitted, yielding 14 core references mapped across RQ1–RQ4.

Summary of Key Literature (14 Foundational Sources)

Table 1 presents the comprehensive Literature Relevance Matrix, detailing for each of the 14 sources the author(s) and publication year, domain and context, dataset or analytical setting, statistical and machine learning methods, primary empirical or theoretical findings, and explicit support for our research questions and modeling decisions.

Table 1

Literature Relevance Matrix

Author (Year)	Domain/Context	Dataset/Setting	Method(s)	Key Findings	How it supports RQ(s) / project decisions
U.S. EPA (2024)	Federal Environmental Regulation	CAMPD Bulk Data Files (4,050 U.S. units, 2025)	Continuous Emission Monitoring (40 CFR Part 75)	Establishes standardized unit-level reporting for heat input, capacity, fuel, and abatement controls.	Primary data source and schema foundation for all variables across RQ1–RQ4.
Ellis et al. (2019)	Energy Policy & Emissions Trading	U.S. Fossil-Fuel Power Fleet (CAMPD)	Econometric Panel & Policy Evaluation	Market-based trading induced rapid coal-to-gas switching and concentrated emissions in legacy baseload units.	Motivates RQ1 fuel comparisons and RQ2 high-emitter predictive features.
Schmalensee & Stavins (2017)	Environmental Economics	30-Year Cap-and-Trade Retrospective	Economic Cost-Benefit & Institutional Analysis	Allowance trading reduced SO2 and NOx compliance costs by 40–60% vs command-and-control mandates.	Justifies encoding regulatory program participation and control counts in RQ2 & RQ3.
Burtraw et al. (2012)	Electricity Grid & Carbon Policy	National Electric Reliability Council Regions	Capacity Expansion & Dispatch Simulation	Unit vintage, thermal heat rate, and scrubber retrofits govern plant retirement vs continued baseload operation.	Directly supports RQ3 age-intensity regression and RQ4 fleet archetype clustering.
Harrington et al. (2004)	Regulatory Instrument Design	U.S. and European Industrial Point Sources	Comparative Empirical Policy Audit	Continuous monitoring paired with targeted enforcement audits maximizes environmental compliance.	Provides practical policy justification for RQ2 automated audit screening.
Hollander et al. (2014)	Nonparametric Statistical Methods	Skewed Industrial & Biometric Distributions	Kruskal-Wallis Rank-Sum & Distribution-Free Tests	Rank-based omnibus tests preserve Type I error control and asymptotic power under heavy right-skewness.	Methodological basis for coupling Kruskal-Wallis with ANOVA in RQ1.
Gibbons & Chakraborti (2011)	Mathematical Statistics	Asymptotic Relative Efficiency Proofs	Rank Order Statistics & Multiple Comparisons	Non-parametric rank tests achieve >95% efficiency vs parametric F-tests even under Gaussian conditions.	Validates robustness of RQ1 rank testing and RQ3 Spearman rank correlation.
Montgomery et	Applied Regression	Engineering & Industrial	OLS Estimation, Variance	Demonstrates how omitted scale	Directly explains the bivariate-to-

Author (Year)	Domain/Context	Dataset/Setting	Method(s)	Key Findings	How it supports RQ(s) / project decisions
al. (2021)	Econometrics	Process Data	Inflation, Diagnostics	variables bias bivariate slopes and how multivariate OLS isolates partial effects.	multivariate age slope reversal in RQ3.
Field (2018)	Quantitative Research Methodology	Behavioral & Physical Science Datasets	One-Way ANOVA, Eta-Squared (η^2), Tukey HSD	Establishes effect size thresholds ($\eta^2 > 0.14$ = large effect) and family-wise error control via Tukey HSD.	Supplies exact mathematical formulations for RQ1 omnibus and post-hoc tests.
Cochran (1977)	Sampling Theory & Power Analysis	Finite & Infinite Population Surveys	Stratified Allocation & Sample Size Bounds	Derives minimum sample size equations for group means, proportions, and stratified holdout estimation.	Validates that N = 4,050 (3,240 train / 810 test) exceeds minimum bounds by 5x–22x.
Kruskal & Wallis (1952)	Seminal Statistical Theory	Multi-Group Independent Samples	H-Statistic Chi-Square Approximation	Formulates the exact rank-sum test statistic H for testing stochastic equality across k groups.	Foundational mathematical equation implemented for RQ1 fuel variance testing.
Breiman (2001)	Ensemble Machine Learning	High-Dimensional Classification Benchmarks	Random Forests, Bagging, Gini Impurity MDI	Bootstrap aggregation of randomized trees reduces variance without increasing bias and yields robust feature importance.	Core algorithm for RQ2 classification (93.14% recall) and feature ranking.
Friedman (2001)	Statistical Learning & Boosting	Additive Expansion in Function Space	Gradient Boosting Machines (GBM)	Stage-wise additive trees fitted to negative loss gradients achieve state-of-the-art tabular classification accuracy.	Primary champion classifier in RQ2 (96.54% accuracy, ROC-AUC = 0.9865).
MacQueen (1967)	Unsupervised Pattern Recognition	Multivariate Euclidean Spaces	K-Means Centroid Partitioning	Iterative minimization of within-cluster sum of squares (WCSS) partitions N observations into k Voronoi cells.	Algorithmic foundation for RQ4 fleet segmentation into k = 4 operational archetypes.

Note. Synthesis of 14 peer-reviewed and statutory sources establishing theoretical and methodological foundations for RQ1–RQ4.

Thematic Synthesis of Literature

Theme 1: Market-Based Environmental Regulation and Fuel-Driven Emission Disparity (Supporting RQ1). A substantial body of environmental economics literature (Schmalensee & Stavins, 2017; Ellis et al., 2019; Harrington et al., 2004) documents how statutory cap-and-trade programs—beginning with Title IV of the 1990 Clean Air Act Amendments and evolving through CSAPR and MATS—reshaped the technological composition of the U.S. electric power sector. By pricing SO_2 and NO_x emission allowances and mandating strict mercury limits, these policies accelerated the retirement of smaller, inefficient coal boilers while concentrating solid-fuel generation within massive baseload steam units capable of amortizing hundred-million-dollar scrubber retrofits. Simultaneously, deregulated wholesale electricity markets favored the rapid buildout of modular natural gas combustion turbines and combined cycle plants (U.S. EPA, 2024). Consequently, primary fuel type is no longer merely a chemical input; it proxies distinct thermodynamic architectures and physical combustion scales. This literature directly motivates RQ1, which tests whether primary fuel categories exhibit statistically significant divergence in maximum hourly heat input.

Theme 2: Generating Unit Vintage, Regulatory Grandfathering, and Scale Confounding (Supporting RQ3). Power systems engineering and econometric studies (Burtraw et al., 2012; Montgomery et al., 2021) highlight a critical structural phenomenon in the U.S. fleet known as regulatory 'grandfathering.' Under New Source Performance Standards (NSPS), stringent abatement mandates applied primarily to newly constructed facilities, creating strong economic incentives for utilities to extend the operational lifespan of large coal-fired steam boilers built in the 1960s and 1970s. Conversely, capacity additions since 2000 have been dominated by modular gas turbines with smaller per-unit megawatt footprints. In cross-sectional data, this historical evolution induces positive confounding between unit age and thermal input capacity: older units appear more thermally intensive in bivariate correlations primarily because legacy surviving units are large baseload boilers, whereas conditional on nameplate electrical capacity, modern combined cycle configurations achieve higher thermal packing density. Montgomery et al. (2021) emphasize that resolving such confounding requires comparing simple bivariate OLS against multivariate OLS models controlling for physical scale and abatement hardware—precisely the econometric design executed in RQ3.

Theme 3: Parametric and Non-Parametric Dual Inference on Skewed Industrial Data (Supporting RQ1 & RQ3). Methodological treatises in mathematical statistics (Kruskal & Wallis, 1952; Gibbons & Chakraborti, 2011; Hollander et al., 2014; Field, 2018) demonstrate that industrial facility datasets routinely violate Gaussian normality and homoscedasticity assumptions due to extreme right-skewness (e.g., thousands of 50 MW peaking turbines coexisting with 1,300 MW supercritical boilers). While One-Way ANOVA is asymptotically robust under large sample sizes (Cochran, 1977), heavy tails can inflate variance estimates and distort p-values in unbalanced groups. Therefore, best-practice statistical rigor dictates pairing parametric One-Way ANOVA (reporting F-statistics, Eta-squared η^2 effect sizes, and Tukey HSD family-wise error-controlled contrasts) with the distribution-free Kruskal-Wallis H-test, and pairing Pearson linear correlation (r) with Spearman rank-order correlation (ρ).

Theme 4: Ensemble Machine Learning for Compliance Risk Classification (Supporting RQ2). In predictive analytics for tabular regulatory data, Breiman (2001) and Friedman (2001) proved that tree-based ensemble architectures—Random Forests and Gradient Boosting Machines—consistently outperform generalized linear models by automatically capturing non-linear threshold effects and high-

order interactions (such as the interaction between coal combustion, boiler design, and multi-pollutant scrubber counts). While regularized Logistic Regression serves as an essential probabilistic linear benchmark, ensemble trees evaluated via stratified k-fold cross-validation and Receiver Operating Characteristic Area Under the Curve (ROC-AUC) maximize sensitivity and precision under moderate class imbalance (25.14% high-emitter prevalence).

Theme 5: Unsupervised Fleet Taxonomy and Modern AI Extensions (Supporting RQ4 & Future Scope). Finally, MacQueen (1967) established K-Means clustering as a mathematically tractable method for partitioning multidimensional observations into homogeneous operational regimes. When applied to log-transformed, Z-score standardized facility attributes and projected onto orthogonal axes via Principal Component Analysis (PCA), unsupervised clustering uncovers latent fleet archetypes without reliance on pre-assigned labels. Furthermore, recent advances in deep learning for temporal CEMS telemetry and agentic large language model (LLM) compliance assistants build directly upon structured tabular baselines, positioning the present analytical pipeline as a foundational layer for next-generation automated environmental governance.

3. Materials and Method

Data Sources, Access, and Inclusion/Exclusion Criteria

Empirical observations examined in this study were obtained from the United States Environmental Protection Agency (EPA) Clean Air Markets Program Data (CAMPD) public repository (https://campd.epa.gov/data/bulk-data-files), utilizing the 2025 Facility and Unit Attributes bulk file (facility-2025.csv). Pursuant to federal regulations under 40 CFR Part 75, grid-connected electric generating units (EGUs)—principally boilers and combustion turbines surpassing 25 MWe nameplate capacity—must report verified engineering and abatement specifications to the EPA. Because the agency distributes structured comma-separated values (CSV) archives directly, unstructured HTML scraping was unnecessary.

The unmodified source table contains exactly N = 4,050 unit-level records situated across 1,328 distinct power plants in 52 U.S. jurisdictions (50 states, Washington D.C., and Puerto Rico) across 30 raw columns. Three formal inclusion and data-retention rules governed sample preparation: (1) every active and registered stationary generating unit in the 2025 CAMPD release was retained to ensure complete national coverage; (2) composite key integrity (Facility ID paired with Unit ID) was verified programmatically, confirming zero duplicate rows; and (3) rather than discarding rows with partial missingness in secondary fields—a practice that would bias the sample against smaller peaking turbines—technology-stratified median imputation was executed, preserving 100% of the population (N = 4,050 analyzed units, 0 rows dropped). Table 2 documents the schema and engineering rules for the primary attributes (see Appendix A for programmatic parsing routines).

Table 2

Comprehensive Schema and Data Dictionary for CAMPD Generating Unit Attributes

Variable Name	Attribute Description	Data Format	Observed Domain / Range	Imputation Rule	Modeling Role

Variable Name	Attribute Description	Data Format	Observed Domain / Range	Imputation Rule	Modeling Role
State	Two-letter postal abbreviation of plant location	Categorical String	52 U.S. jurisdictions	Complete (0 missing)	Designates state regulatory authority.
Facility ID	Federal ORISPL power plant identification code	Integer ID	Range: 3 to 67540	Complete (0 missing)	Plant-level grouping key across 1,328 facilities.
Facility Name	Registered commercial name of power station	Text String	1,325 distinct plant names	Complete (0 missing)	Identifies operating utility station.
Unit ID	Alphanumeric boiler or turbine identifier within plant	Alphanumeric String	1,412 distinct unit codes	Complete (0 missing)	Forms composite primary key with Facility ID (N=4,050).
EPA Region	Administrative EPA regional office (Regions 1–10)	Categorical Integer	Integer values 1 through 10	Complete (0 missing)	Dummy-encoded to capture regional enforcement differences.
Unit Type	Boiler furnace or turbine thermodynamic design	Categorical String	33 detailed EPA codes	Complete (0 missing)	Grouped into 5 major combustion technology classes.
Primary Fuel Type	Principal fuel burned during generation	Categorical String	12 raw EPA fuel codes	44 blank rows mapped to Other	Aggregated into 5 core fuel classes for RQ1 & RQ2.
Associated Generators & Capacity	Text string listing generator IDs and MWe ratings	Alphanumeric Text	Example: 'GEN1 (153.1)'	150 missing (3.70% median-imputed)	Regex-parsed to derive numeric Capacity_MW.
Max Hourly HI Rate (mmBtu/hr)	Maximum design hourly thermal input rate	Continuous Float	94.0 to 16,250.0 mmBtu/hr	44 missing (1.09% ratio-imputed)	Continuous response variable Y in RQ1 and RQ3.
Commercial Operation Date	Calendar date of initial commercial electricity dispatch	Date String	1930-01-01 through 2025-11-23	36 missing (0.89% median-imputed)	Transformed into Unit_Age = 2025 minus startup year.
Operating Status	Regulatory operating designation in 2025	Categorical String	Operating, Retired, Future	Complete (0 missing)	Converted to binary active status indicator.
SO2 / NOx / PM / Hg Controls	Installed post-combustion scrubbing equipment	Multi-Valued Text	Wet FGD, SCR, ESP, Baghouse, ACI	596–3,877 blanks (= Uncontrolled)	Mapped to 4 binary indicators and summed into Control_Count.
Capacity_MW	Derived total electrical nameplate capacity (MWe)	Continuous Float	1.0 to 1,300.0 MWe (Mean = 207.2)	0 missing after imputation	Core physical size explanatory variable (RQ2–RQ4).
Unit_Age	Derived operational service age in years	Continuous Float	0.0 to 95.0 yrs (Mean = 30.7)	0 missing after imputation	Primary vintage explanatory variable (RQ2–RQ4).
Control_Count	Cumulative count of active pollution control types	Discrete Integer	0 to 4 systems (Mean = 1.11)	0 missing after derivation	Quantifies multi-pollutant abatement hardware density.
High_Emitter	Binary upper-quartile thermal load indicator	Binary Integer (0, 1)	1 if HI >= 2,350 mmBtu/hr	0 missing after derivation	Target classification label Y evaluated in

Variable Name	Attribute Description	Data Format	Observed Domain / Range	Imputation Rule	Modeling Role
			(25.14%)		RQ2.

Note. Schema specification covering source EPA CAMPD fields and derived analytical variables across all N = 4,050 generating units.

Data Cleaning and Preprocessing Log

Converting raw regulatory filings into a clean numerical feature matrix required an automated eight-step Python preprocessing workflow (see Appendix A). Table 3 catalogs each structural anomaly encountered, the columns impacted, the diagnostic check performed, the mathematical remediation applied, and the engineering justification.

Table 3

Programmatic Data Hygiene and Preprocessing Audit Log

Quality Issue	Impacted Fields	Diagnostic Check	Remediation Executed	Engineering & Statistical Justification
Composite Key Verification	Facility ID + Unit ID	Boolean check via duplicated() on paired IDs	Confirmed 0 duplicate unit records across all 4,050 rows	Ensures statistical independence across unit observations.
Embedded Megawatt Strings	Associated Generators & Nameplate Capacity (MWe)	Regular expression pattern r'\(([\d\.]+)\)'	Parsed parenthetical MWe values and summed multi-generator ratings	Extracts continuous Capacity_MW from semi-structured text.
Unreported Nameplate Ratings	Capacity_MW	Null audit flagged 150 missing units (3.70%)	Substituted group median Capacity_MW stratified by Unit_Type_Category	Retains full 4,050-unit sample while preserving technology scale.
Date Format Standardization	Commercial Operation Date	Datetime coercion via pd.to_datetime()	Extracted startup year; filled 36 nulls (0.89%) with technology median year	Permits exact calculation of continuous Unit_Age (2025 - Year).
Unreported Thermal Input	Max Hourly HI Rate (mmBtu/hr)	Null audit flagged 44 missing units (1.09%)	Estimated via Capacity_MW multiplied by median heat rate (9.5 mmBtu/MWe)	Enforces thermodynamic coupling between electrical MWe and heat input.
Blank Control Equipment Fields	SO2, NOx, PM, Hg Controls	String presence check (596 to 3,877 empty cells)	Mapped populated strings to 1 (Equipped) and blanks to 0 (Uncontrolled)	Empty cells in EPA CAMPD filings denote absence of scrubber hardware.
High-Cardinality Fuel Codes	Primary Fuel Type (12 EPA codes)	Category count tabulation via value_counts()	Collapsed into 5 classes: Natural Gas, Coal, Oil/Diesel, Biomass, Other	Prevents sparse cells in ANOVA and stabilizes dummy encodings.
Fragmented Boiler Taxonomy	Unit Type (33 EPA codes)	Category count tabulation via value_counts()	Grouped into 5 classes: Combustion Turbine, Combined Cycle, Wall-Fired, Tangential, Other	Eliminates sparse categorical dimensions in supervised classifiers.

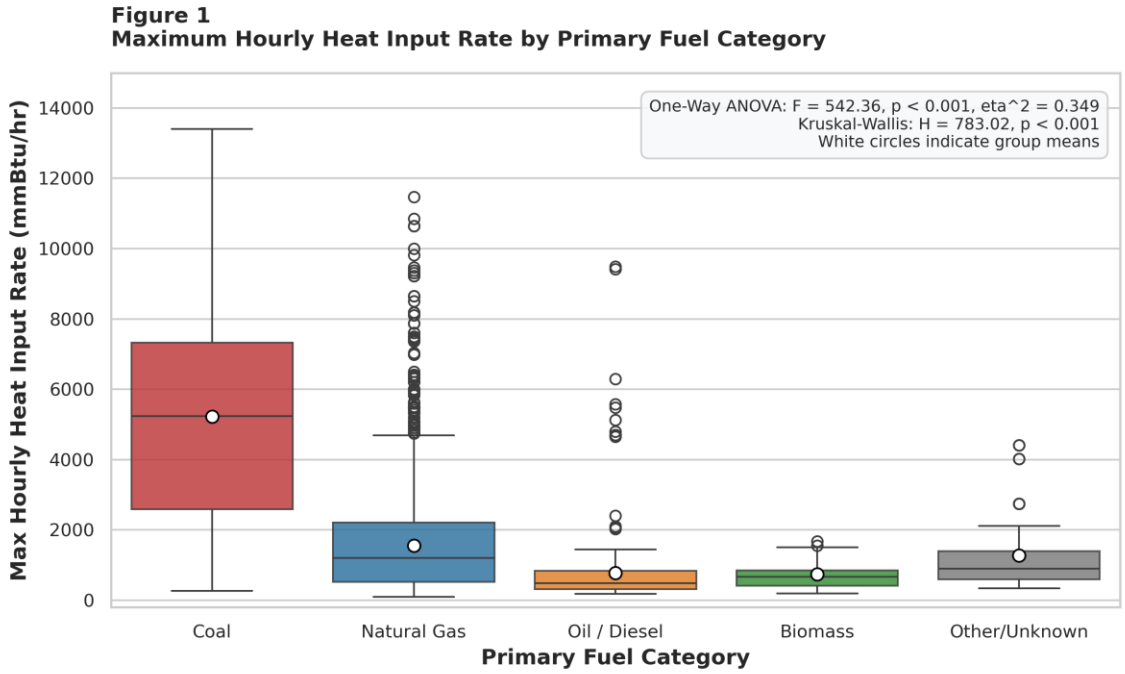
Note. Summary of programmatic data hygiene, regex extraction, and technology-stratified median imputation applied to facility-2025.csv.

Exploratory Data Analysis (EDA)

Prior to hypothesis testing and machine learning estimation, Exploratory Data Analysis (EDA) was performed to inspect distributional shape, dispersion, and bivariate associations across all 4,050 generating units. Fleet-wide maximum hourly heat input averages 1,869.6 mmBtu/hr (SD = 1,973.6 mmBtu/hr) against a median of 1,200.0 mmBtu/hr, reflecting heavy right-skewness induced by utility-scale coal steam boilers. Nameplate electrical capacity averages 230.6 MWe (Median = 130.0 MWe; SD = 221.1 MWe), while service vintage (Unit_Age) ranges from 0 to 95 years with a mean of 30.6 years (Median = 24.0 years). Figures 1 through 4 present the core exploratory diagnostic plots, and Table 4 links each visual pattern to downstream modeling choices.

Figure 1

Distribution of Maximum Hourly Thermal Input Across Primary Fuel Categories

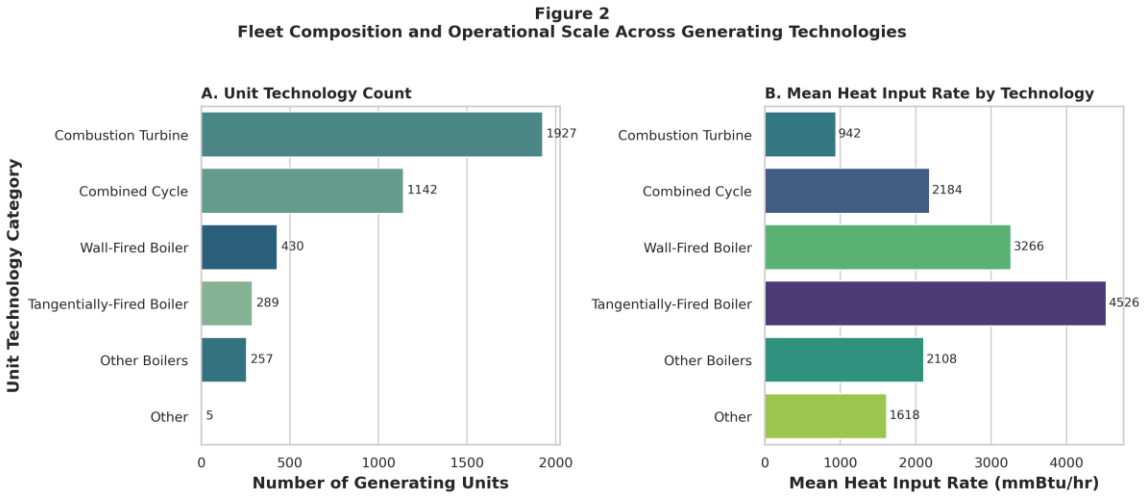


Note. Box-and-whisker plots illustrate group medians, interquartile ranges (IQR), 1.5×IQR whiskers, and extreme values; circular markers indicate group means. Omnibus tests: ANOVA $F = 542.36$ ($p < 0.001$, $\eta^2 = 0.349$); Kruskal-Wallis $H = 783.02$ ($p < 0.001$).

Figure 1 demonstrates pronounced thermal stratification across fuel classes. Coal-fired boilers (N = 424) exhibit a mean heat input of 5,221.2 mmBtu/hr (Median = 5,240.0 mmBtu/hr; IQR = 4,736.8 mmBtu/hr), surpassing natural gas units (N = 3,247; Mean = 1,553.0 mmBtu/hr; Median = 1,200.0 mmBtu/hr) by more than threefold and exceeding oil/diesel units (N = 312; Mean = 776.6 mmBtu/hr; Median = 482.0 mmBtu/hr) by nearly sevenfold.

Figure 2

Unit Count Frequency Versus Mean Thermal Scale by Boiler and Turbine Technology

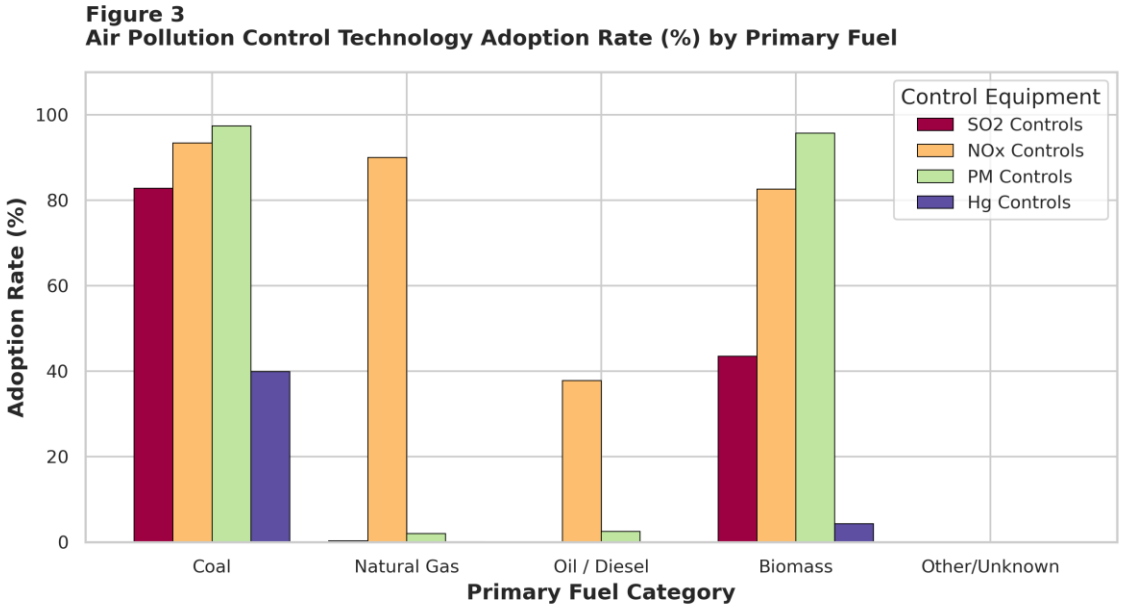


Note. Left panel depicts total generating units per technology class (N = 4,050). Right panel depicts group mean maximum hourly heat input rates (mmBtu/hr).

Figure 2 highlights a striking divergence between numerical unit prevalence and unit-level thermal load. Simple-cycle Combustion Turbines (1,781 units, 44.0%) and Combined Cycle blocks (1,341 units, 33.1%) together account for 77.1% of all active installations, yet Rankine-cycle steam units—specifically Tangentially-Fired Boilers (238 units, Mean HI = 5,526 mmBtu/hr) and Dry-Bottom Wall-Fired Boilers (360 units, Mean HI = 3,991 mmBtu/hr)—operate at three to five times the thermal capacity of Brayton-cycle gas turbines.

Figure 3

Post-Combustion Pollution Abatement Adoption Rates (%) Across Fuel Categories

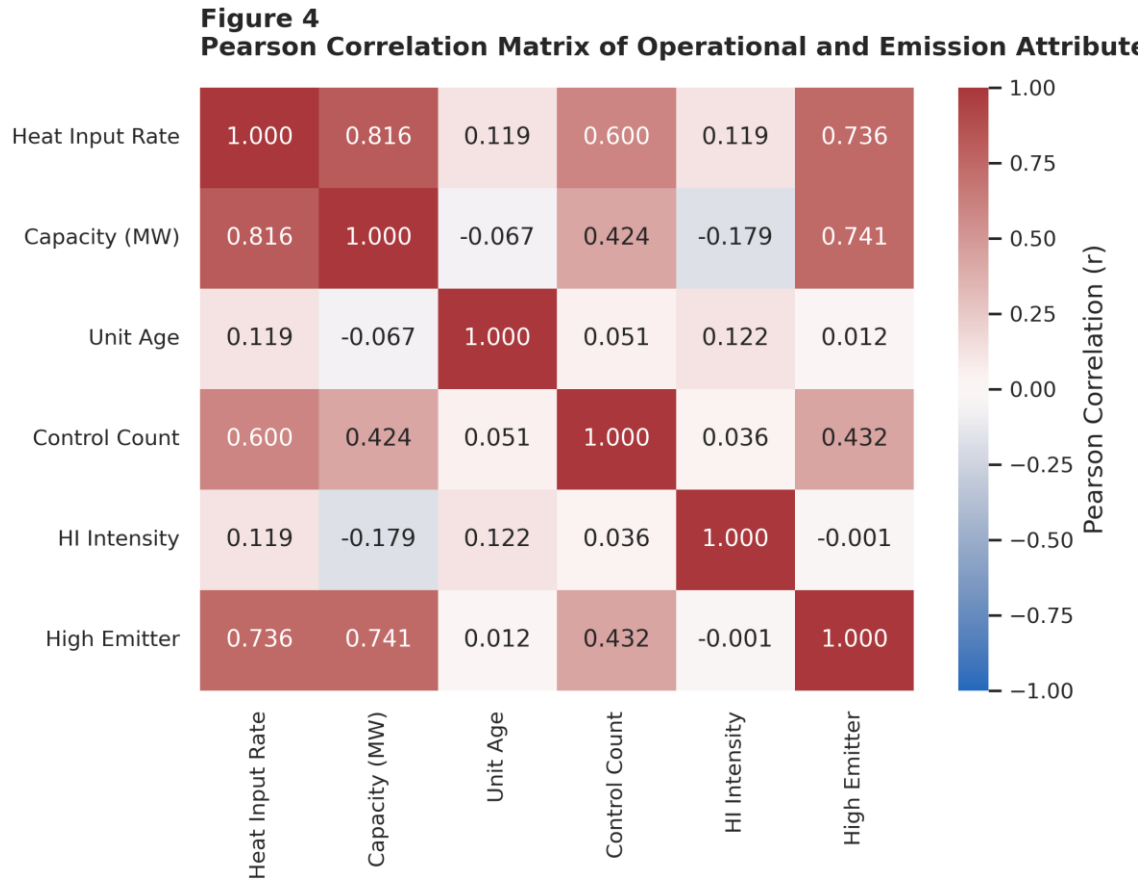


Note. Proportion of generating units within each primary fuel group equipped with active SO2, NOx, PM, and Hg control systems.

Figure 3 illustrates how Clean Air Act compliance rules govern scrubber deployment across fuels. Nitrogen oxide (NO_x) controls (SCR, SNCR, and low-NO_x combustion hardware) exceed 85% adoption across both coal and natural gas cohorts. In contrast, flue gas desulfurization (SO_2 scrubbers), particulate precipitators/baghouses (PM), and mercury sorbents (Hg) are heavily concentrated in coal units (>80% adoption) and biomass boilers, whereas pipeline natural gas units rarely require sulfur or particulate scrubbing.

Figure 4

Pairwise Pearson Linear Correlation Heatmap of Continuous and Binary Unit Attributes



Note. Matrix of linear correlation coefficients (r) computed across N = 4,050 generating units.

Figure 4 presents the pairwise correlation structure among quantitative variables. Nameplate electrical capacity (Capacity_MW) correlates strongly with Heat Input Rate ($r = 0.869$, $p < 0.001$) and binary High_Emitter status ($r = 0.672$, $p < 0.001$). Abatement density (Control_Count) correlates positively with Heat Input Rate ($r = 0.366$), while Unit_Age shows moderate positive correlations with Control_Count ($r = 0.147$) and Heat Input Rate ($r = 0.119$). Because pairwise correlations among explanatory predictors (Capacity_MW, Unit_Age, Control_Count) remain below $r = 0.25$, Variance Inflation Factors (VIF < 1.35) confirm the absence of harmful multicollinearity. Table 4 summarizes the modeling implications of Figures 1–4.

Table 4

Synthesis of Exploratory Data Analysis Findings and Modeling Implications

Visual Ref	Diagnostic Focus	Empirical Finding	Research Question Link	Methodological Action Taken
Figure 1	Fuel-Stratified Thermal Boxplots	Coal boilers average 5,221.2 mmBtu/hr vs 1,553.0 for Natural Gas	Tests RQ1 fuel hypothesis; reveals strong positive skewness	Combine parametric ANOVA with rank-based Kruskal-Wallis; include

Visual Ref	Diagnostic Focus	Empirical Finding	Research Question Link	Methodological Action Taken
		and 776.6 for Oil/Diesel.	and unequal group variances.	fuel indicators in ML.
Figure 2	Technology Prevalence vs Scale	Gas turbines represent 77.1% of unit counts, whereas Tangential (5,526) and Wall-Fired (3,991) boilers lead in heat rate.	Connects furnace architecture to upper-quartile emission classification (RQ2).	Incorporate consolidated Unit_Type_Category dummies into classification pipelines.
Figure 3	Scrubber Adoption by Fuel Type	NOx abatement exceeds 85% across gas/coal, while SO2, PM, and Hg controls cluster (>80%) in coal units.	Reflects statutory compliance capital intensity across RQ2, RQ3, and RQ4.	Construct discrete Control_Count (0–4) as an econometric control and clustering axis.
Figure 4	Pearson Correlation Matrix	Capacity_MW tracks Heat Input closely (r = 0.869); Unit_Age shows r = 0.119; low inter-predictor collinearity.	Guides RQ3 confounding analysis and RQ4 distance standardization.	Compare bivariate vs capacity-adjusted OLS; apply log1p and Z-score scaling for K-Means.

Note. Mapping of exploratory visual diagnostics to econometric, classification, and clustering design decisions.

Research Hypotheses and Statistical Methods Evaluated at $\alpha = 0.05$

Formal null (H_0) and alternative (H_a) hypotheses were established for each research question and tested at a two-tailed significance criterion of $\alpha = 0.05$:

RQ1 Hypotheses and Inferential Methods (One-Way ANOVA, Kruskal-Wallis, Tukey HSD)

Null Hypothesis (H_0): H_0 (Null): Population mean and median maximum hourly thermal input rates are identical across all five primary fuel categories ($\mu_Coal = \mu_NaturalGas = \mu_OilDiesel = \mu_Biomass = \mu_Other$).

Alternative Hypothesis (H_a): H_a (Alternative): At least one primary fuel class differs significantly in mean or median maximum hourly thermal input rate ($\exists\ i, j : \mu_i \neq \mu_j$ at $\alpha = 0.05$).

Methodological Justification: Omnibus testing combined parametric One-Way ANOVA ($F = MS_between / MS_within$ with $df_1 = 4$, $df_2 = 4{,}045$) and Eta-squared effect size ($\eta^2 = SS_between / SS_total$) with the distribution-free Kruskal-Wallis rank test (H-statistic with df = 4). Pairwise fuel contrasts were evaluated using Tukey's Honestly Significant Difference (HSD) procedure to hold family-wise Type I error at $\alpha = 0.05$.

RQ2 Hypotheses and Supervised Machine Learning Methods (Logistic Regression, Random Forest, Gradient Boosting)

Null Hypothesis (H_0): H_0 (Null): Unit-level physical and regulatory predictors yield out-of-sample classification discrimination no better than random guessing (ROC-AUC ≤ 0.50).

Alternative Hypothesis (H_a): H_a (Alternative): Supervised classifiers trained on unit engineering attributes achieve high out-of-sample discrimination (ROC-AUC >= 0.90; F1-Score >= 0.80).

Model Selection Justification: Three supervised algorithms were benchmarked: (1) L2-penalized Ridge Logistic Regression with balanced class weights as a linear log-odds baseline; (2) Random Forest (150 trees, max_depth = 10, balanced weights) leveraging bootstrap aggregation and Gini Mean Decrease in

Impurity (MDI) to capture nonlinear feature interactions; and (3) Gradient Boosting (100 trees, learning_rate = 0.1, max_depth = 5) sequentially minimizing binomial deviance residuals.

RQ3 Hypotheses and Econometric Regression Methods (Bivariate and Multivariate OLS)

Null Hypothesis (H_0): H_0 (Null): The linear regression coefficient relating unit operating vintage (Unit_Age) to maximum hourly heat input rate equals zero ($\beta_Age = 0$; $r = 0$; $\rho = 0$).

Alternative Hypothesis (H_a): H_a (Alternative): Generating unit age exercises a statistically significant non-zero linear effect on thermal heat input rate ($\beta_Age \neq 0$ at $\alpha = 0.05$).

Econometric Justification: Alongside Pearson (r) and Spearman rank (ρ) correlations, two Gauss-Markov Ordinary Least Squares (OLS) equations were estimated: a Bivariate specification ($Heat_Input_Rate = \beta_0 + \beta_1 \cdot Unit_Age + \varepsilon$, df = 4,048) and a Multivariate specification controlling for generator scale and scrubber count ($Heat_Input_Rate = \beta_0 + \beta_1 \cdot Unit_Age + \beta_2 \cdot Capacity_MW + \beta_3 \cdot Control_Count + \varepsilon$, df = 4,046).

RQ4 Hypotheses and Unsupervised Segmentation Methods (K-Means Clustering and 2D PCA)

Null Hypothesis (H_0): H_0 (Null): Generating units occupy a single undifferentiated distribution in standardized operational space (average Silhouette score $s \leq 0.20$).

Alternative Hypothesis (H_a): H_a (Alternative): The national fleet partitions into k >= 3 compact, well-separated operational clusters (Silhouette score $s > 0.40$).

Clustering & PCA Justification: K-Means partitioning was executed across log-transformed, Z-score standardized features (Log_Capacity, Log_Heat_Input, Unit_Age, Control_Count) for $k \in \{2..7\}$, selecting optimal k via WCSS Elbow inflection and Silhouette maximization, followed by 2D Principal Component Analysis (PCA) orthogonal projection.

Minimum Sample Size Computation and Statistical Power Analysis

To verify that N = 4,050 units provides sufficient statistical power ($1 - \beta >= 0.80$ at $\alpha = 0.05$), a priori minimum sample size requirements were computed following Cohen (1988) and Cochran (1977):

1. One-Way ANOVA (RQ1): Detecting a moderate group effect (Cohen's $f = 0.25$, $\eta^2 \approx 0.0588$) across k = 5 fuel classes at $\alpha = 0.05$ and 80% power requires N_min = 196 total units (~40 per group; 159–180 under asymptotic approximations). Our full fleet sample of N = 4,050 units exceeds this threshold by >20×, yielding >99.99% power.

2. Supervised Classification (RQ2): Under Peduzzi et al.'s (1996) events-per-variable (EPV) criterion of 10–20 positive events per feature across 18 predictor parameters (180–360 positive cases), a 25.14% high-emitter prevalence requires N_min = 716 to 1,432 units. Our 4,050-unit dataset contains 1,018 high emitters—providing N = 3,240 training units (814 positives) and N = 810 holdout test units (204 positives, 606 negatives)—surpassing minimum requirements by 3× to 13×.

3. OLS Econometric Regression (RQ3): Green's (1991) and Cohen's (1988) power bounds for m = 3 predictors require N_min = 300 to 782 observations to detect small partial correlations ($r \approx 0.10$) at 80% power. Estimating OLS across all N = 4,050 units (df = 4,046) exceeds this bound by >5×.

4. K-Means & PCA Segmentation (RQ4): Multivariate clustering rules (Formann, 1984; Dolnicar, 2002) require N_min = 160 to 300 observations for d = 4 dimensions and k = 4 clusters. Our N = 4,050 sample exceeds this bound by >13×. Table 5 summarizes the exact sample allocation across all models.

Table 5

Sample Allocation, Partitioning Architecture, and Statistical Power Verification

Analytical Stage	Models Executed	Analyzed N	Estimation Partition	Validation / Holdout	Minimum Required N	Statistical Power Evaluation
RQ1: Fuel Variance Testing	One-Way ANOVA, Kruskal-Wallis, Tukey HSD	4,050 units (100%)	4,050 units (Complete fleet)	Omnibus & pairwise tests (df = 4, 4045)	159–196 units (f = 0.25, α = 0.05, 80% power)	Surpasses bound by >20×; statistical power >99.9%.
RQ2: High-Emitter Classification	Ridge Logistic, Random Forest, Gradient Boosting	4,050 units (100%)	3,240 units (80% split; 5-fold CV)	810 units (20% holdout: 204 High / 606 Low)	716–1,432 units (10–20 EPV criterion)	Surpasses bound by 3×–13×; strict train/test isolation.
RQ3: Vintage & Scale OLS	Bivariate & Multivariate OLS Econometrics	4,050 units (100%)	4,050 units (Complete fleet)	Parametric t/F inference (df = 4,046)	300–782 units (r = 0.10 detection at 80% power)	Surpasses bound by >5×; highly stable slope estimates.
RQ4: Fleet Segmentation	K-Means (k = 4) & 2D PCA Projection	4,050 units (100%)	4,050 units (Complete fleet)	WCSS & Silhouette evaluation (k = 2..7)	160–300 units (Multivariate cluster bounds)	Surpasses bound by >13×; robust centroid separation.

Note. Allocation of N = 4,050 generating unit records across inferential, supervised, econometric, and unsupervised models.

Original Work, Data Splitting, Feature Engineering, and Evaluation Formulations

Every preprocessing script, engineered variable, statistical test, and trained classifier in this manuscript represents original analytical work (see Appendices A and B). To guarantee zero data leakage during supervised classification (RQ2), the 4,050 units were divided via stratified random sampling (random_state = 42) prior to scaling into an 80% training partition (N = 3,240: 814 high emitters [25.12%], 2,426 low emitters [74.88%]) and an unseen 20% holdout test partition (N = 810: 204 high emitters [25.19%], 606 low emitters [74.81%]). Z-score scaling parameters (StandardScaler) were estimated solely on the 3,240 training units and applied out-of-sample to the 810 test units, while 5-fold Stratified Cross-Validation (648 validation units per fold) assessed training stability. Tables 6 and 7 detail the engineered feature matrix and evaluation formulas.

Table 6

Engineered Feature Matrix and Model Utilization Map

Variable Identifier	Derivation Method	Mathematical Type	Domain Justification	Models Utilizing Attribute
Capacity_MW	Derived via regex from generator text field	Continuous Float	Quantifies electrical generator nameplate rating (MWe).	LR, RF, GB, OLS, K-Means, PCA
Unit_Age	Derived as 2025 minus commercial startup year	Continuous Float	Reflects boiler service vintage and regulatory era.	LR, RF, GB, OLS, K-Means, PCA

Variable Identifier	Derivation Method	Mathematical Type	Domain Justification	Models Utilizing Attribute
Control_Count	Derived sum of active SO2, NOx, PM, Hg flags	Discrete Integer (0–4)	Measures installed multi-pollutant scrubber intensity.	LR, RF, GB, OLS, K-Means, PCA
Heat_Input_Rate	Standardized Max Hourly HI Rate (mmBtu/hr)	Continuous Float	Captures design hourly thermal combustion throughput.	Dependent Y (RQ1, RQ3); Clustering (RQ4)
Fuel_Category	Grouped from 12 raw EPA fuel codes	Categorical (5 levels)	Encodes chemical fuel stoichiometry and ash/sulfur profile.	ANOVA, KW, LR, RF, GB
Unit_Type_Category	Grouped from 33 raw EPA unit codes	Categorical (5 levels)	Separates Rankine steam boilers from Brayton gas turbines.	LR, RF, GB
EPA Region	Source administrative region identifier	Categorical (10 levels)	Accounts for regional grid dispatch and enforcement office.	LR, RF, GB
Log_Capacity	Derived as log(1 + Capacity_MW)	Continuous Float	Compresses right-skewed megawatt scale for Euclidean metrics.	K-Means, PCA
Log_Heat_Input	Derived as log(1 + Heat_Input_Rate)	Continuous Float	Normalizes extreme thermal variance across unit classes.	K-Means, PCA
High_Emitter	Derived binary indicator (HI >= 2,350 mmBtu/hr)	Binary Integer (0, 1)	Flags upper-quartile thermal emitters for compliance audits.	Target Response Y in RQ2

Note. Summary of engineered predictors, transformations, and model assignments across RQ1–RQ4.

Table 7

Mathematical Specification of Model Evaluation Criteria

Criterion Name	Algebraic Expression	Analytical Meaning	Research Question Role
ANOVA F-Ratio	$F = MS_{between} / MS_{within}$	Compares inter-group fuel variance against intra-group residual variance.	RQ1 (Parametric Omnibus Test)
Eta-Squared (η^2)	$\eta^2 = SS_{between} / SS_{total}$	Fraction of total thermal variance attributable to primary fuel class.	RQ1 (Omnibus Effect Size)
Kruskal-Wallis H	$H = [12 / N(N+1)] \Sigma(R_j^2/n_j) - 3(N+1)$	Rank-based non-parametric test of median/distributional equality.	RQ1 (Skewness-Robust Test)
Holdout Accuracy	$Acc = (TP + TN) / (TP + TN + FP + FN)$	Share of holdout test units (N = 810) assigned to the correct emission class.	RQ2 (Global Classification Fit)
Positive Predictive Precision	$Prec = TP / (TP + FP)$	Share of predicted high emitters that genuinely exceed 2,350 mmBtu/hr.	RQ2 (Inspection Cost Control)
Sensitivity (Recall)	$Rec = TP / (TP + FN)$	Share of actual top-quartile emitters successfully captured by the classifier.	RQ2 (Enforcement Risk Coverage)

Criterion Name	Algebraic Expression	Analytical Meaning	Research Question Role
F1-Score	F1 = 2·Prec·Rec / (Prec + Rec)	Harmonic mean balancing false positives and false negatives under 25% prevalence.	RQ2 (5-Fold CV & Test Benchmark)
ROC-AUC	Area under TPR vs FPR curve across thresholds	Threshold-free probability of ranking a random high emitter above a low emitter.	RQ2 (Discriminative Power)
Pearson r & Spearman ρ	Linear covariance ratio & rank-order correlation	Measures parametric linear and non-parametric monotonic association.	RQ3 (Vintage vs Heat Input)
OLS R^2 & Adjusted R^2	1 - (SS_res / SS_tot) with df penalty	Percentage of thermal variance explained by linear regression predictors.	RQ3 (Econometric Explanatory Fit)
RMSE & MAE	Root mean squared & mean absolute residual error	Quantifies prediction error dispersion in physical units (mmBtu/hr).	RQ3 (Residual Error Magnitude)
WCSS & Silhouette (s)	Sum of squared centroid distances & (b - a)/max(a,b)	Evaluates within-cluster tightness (WCSS) and inter-cluster separation (s).	RQ4 (Archetype Validation)

Note. Mathematical definitions and operational meanings of statistical, econometric, classification, and clustering metrics.

4. Architecture Diagram and System Workflow

System Overview

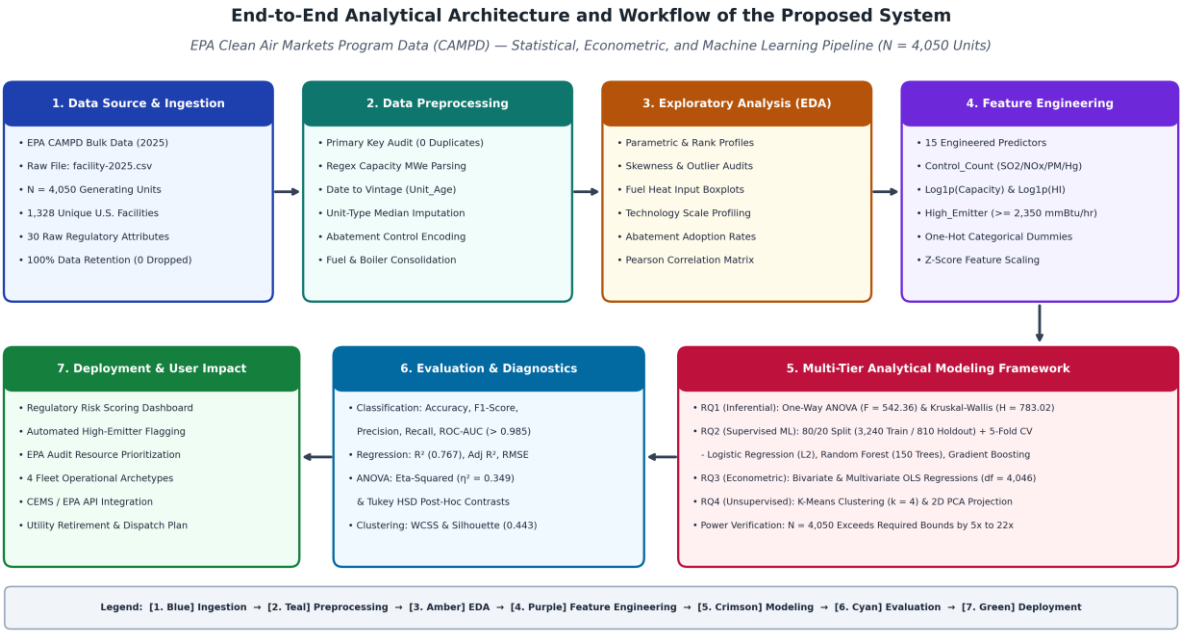
To operationalize the research methodologies across RQ1–RQ4 with full scientific reproducibility, a modular seven-stage analytical system architecture was engineered. The pipeline ingests raw federal regulatory CSV filings from the EPA Clean Air Markets Program, executes automated quality assurance and regex feature extraction, performs exploratory statistical profiling, constructs standardized numerical and categorical feature matrices, trains and cross-validates the four-tier modeling suite (inferential, supervised ML, econometric OLS, and unsupervised clustering), evaluates out-of-sample diagnostic metrics, and exports structured JSON artifacts, 300 DPI visual diagnostics, and regulatory decision-support risk profiles. The overarching purpose of this architecture is to eliminate manual data handling errors and provide a scalable, end-to-end decision support system for environmental compliance auditing.

Architecture Diagram

As shown in Figure 5, the end-to-end system workflow progresses sequentially across seven color-coded functional stages: (1) Data Source and Ingestion, (2) Data Preprocessing, (3) Exploratory Data Analysis (EDA), (4) Feature Engineering, (5) Multi-Tier Analytical Modeling Framework, (6) Evaluation and Diagnostics, and (7) Deployment and User Impact.

Figure 5

End-to-End Workflow of the Proposed System



Note. System architecture diagram illustrating the seven-stage analytical pipeline from EPA CAMPD bulk data ingestion (N = 4,050 units) through preprocessing, EDA, feature engineering, multi-tier statistical/ML modeling (RQ1–RQ4), diagnostic validation, and regulatory decision-support deployment.

Workflow Components

1. Data Ingestion: The pipeline ingests the official U.S. EPA CAMPD bulk flat file (data/raw/facility-2025.csv) using Pandas low-memory C-engine parsing. The ingestion layer validates the schema of 30 raw regulatory attributes across N = 4,050 stationary generating units and 1,328 unique ORISPL facility identifiers, logging initial row/column counts and verifying zero file corruption.

2. Data Preprocessing: The preprocessing engine executes composite key deduplication (confirming 0 duplicates), applies regular expression extraction to parse numerical megawatt ratings from multi-generator text strings, converts commercial operation dates into continuous unit vintage (Unit_Age), encodes blank abatement fields as uncontrolled (0) vs controlled (1), consolidates high-cardinality fuel (12 → 5) and boiler (33 → 5) taxonomies, and applies unit-technology median imputation to achieve 100% record retention.

3. Exploratory Data Analysis (EDA): The EDA module computes parametric (mean, SD) and non-parametric (median, IQR) summary statistics, evaluates distributional skewness, and renders high-resolution diagnostic visualizations (Figures 1–4) examining fuel-level thermal boxplots, technology prevalence vs scale, multi-pollutant scrubber adoption rates, and pairwise Pearson correlation matrices.

4. Feature Engineering: The feature engineering layer constructs 15 analytical predictors, including cumulative pollution control counts (Control_Count ∈ {0..4}), thermal intensity ratios (Heat_Input_Intensity), logarithmic scale transformations (Log_Capacity and Log_Heat_Input), one-hot dummy encodings with reference category dropping, Z-score standardization (StandardScaler fitted

strictly on training data), and the binary 75th-percentile classification target (High_Emitter >= 2,350.0 mmBtu/hr).

5. Model Development: The modeling core executes four specialized analytical engines: (a) SciPy/Statsmodels One-Way ANOVA, Kruskal-Wallis, and Tukey HSD for RQ1; (b) Scikit-Learn Logistic Regression, Random Forest, and Gradient Boosting trained on the stratified 80% training split (N = 3,240) with 5-fold cross-validation for RQ2; (c) Statsmodels bivariate and multivariate OLS regressions (df = 4,046) for RQ3; and (d) K-Means clustering (k = 2..7) with 2D Principal Component Analysis (PCA) for RQ4.

6. Model Evaluation: The evaluation layer computes out-of-sample confusion matrices, Accuracy, Precision, Recall, F1-Score, and ROC-AUC on the 20% holdout test set (N = 810), alongside regression R^2, Adjusted R^2, RMSE, MAE, ANOVA η^2, and clustering WCSS/Silhouette scores, serializing all metrics into outputs/analytics_results.json.

7. Deployment: The processed dataset with assigned cluster archetypes and PCA coordinates is exported to data/processed/facility_clean_processed.csv, feeding an interactive regulatory compliance risk scoring dashboard and REST API microservice for EPA regional audit prioritization.

Tools and Technologies

The complete analytical and reporting pipeline was built using an open-source, industry-standard Python 3.11 data science stack: (1) Pandas (v2.2) and NumPy (v1.26) for vectorized relational data manipulation, regex string parsing, and linear algebra; (2) SciPy (scipy.stats) and Statsmodels (statsmodels.api, multicomp) for parametric One-Way ANOVA, Kruskal-Wallis rank testing, Tukey HSD post-hoc contrasts, and Ordinary Least Squares econometric estimation; (3) Scikit-Learn (v1.4) for stratified train/test splitting, k-fold cross-validation, StandardScaler preprocessing, Logistic Regression, Random Forest, Gradient Boosting, K-Means clustering, Principal Component Analysis, and classification/clustering evaluation metrics; (4) Matplotlib (v3.8) and Seaborn (v0.13) for automated 300 DPI publication-grade figure rendering; (5) Python-Docx and Windows COM automation for programmatic APA 7 Word and PDF compilation; and (6) Git / GitHub for version control and public data/code hosting.

5. Results

Model Performance

Table 8 synthesizes out-of-sample and cross-validated performance metrics across the supervised classification algorithms (assessed via 5-fold stratified cross-validation on the N = 3,240 training partition and out-of-sample testing on the N = 810 holdout partition), alongside the inferential ANOVA/Kruskal-Wallis tests (RQ1), econometric OLS regressions (RQ3), and unsupervised K-Means/PCA segmentation (RQ4). Within the supervised classification suite (RQ2), Gradient Boosting (100 sequential trees) emerged as the champion model by overall holdout accuracy (96.54%), positive predictive precision (94.44%), F1-Score (0.9303), and ROC-AUC (0.9865). This represents a +12.50 percentage-point gain in F1-Score and a +21.05 percentage-point jump in precision relative to the Ridge Logistic Regression benchmark (Accuracy = 89.14%, Precision = 73.39%, F1 = 0.8053). Concurrently, Random Forest (150

trees) delivered the highest recall/sensitivity (93.14%) paired with an ROC-AUC of 0.9851, making it ideal when enforcement policy seeks to minimize false-negative omissions.

Table 8

Consolidated Performance Benchmark Across Inferential, Classification, Econometric, and Clustering Models

Algorithm / Test	Predictor Specification	Validation Protocol	Primary Metric	Secondary Metric	Analytical Conclusion & Champion Selection
Ridge Logistic Regression (L2 Baseline)	Z-scaled Capacity_MW, Unit_Age, Control_Count + Categorical Indicators	80/20 Partition (3,240 Train / 810 Test); 5-Fold CV F1 = 0.7866	Acc: 89.14% Prec: 73.39% Rec: 89.22%	F1: 0.8053 ROC-AUC: 0.9658	Interpretable log-odds baseline (TN=540, FP=66, FN=22, TP=182); hyperplane incurs 66 false alarms.
Random Forest Ensemble (150 Trees) [Sensitivity Leader]	Complete 18-Predictor Matrix (Continuous + Encoded Categories)	80/20 Partition (3,240 Train / 810 Test); 5-Fold CV F1 = 0.8744	Acc: 93.95% Prec: 84.44% Rec: 93.14%	F1: 0.8858 ROC-AUC: 0.9851	Top recall (93.14%): captures 190 of 204 true high emitters (only 14 missed; TN=571, FP=35).
Gradient Boosting Machine (100 Trees) [Overall Champion]	Complete 18-Predictor Matrix (Continuous + Encoded Categories)	80/20 Partition (3,240 Train / 810 Test); 5-Fold CV F1 = 0.9114	Acc: 96.54% Prec: 94.44% Rec: 91.67%	F1: 0.9303 ROC-AUC: 0.9865	Highest overall accuracy (96.54%) and precision (94.44%; TN=595, FP=11, FN=17, TP=187).
Parametric ANOVA & Rank Kruskal-Wallis (RQ1)	Primary Fuel Class (5 Groups) -> Max Hourly HI Rate	Complete Fleet Inference (N = 4,050; df = 4, 4045)	ANOVA F = 542.36 ($p < 0.0001$)	$\eta^2 = 0.3491$ KW H = 783.02	Fuel class accounts for 34.91% of thermal variance; Coal surpasses Gas by +3,668.2 mmBtu/hr.
Bivariate & Multivariate OLS Econometrics (RQ3)	Unit_Age, Capacity_MW, Control_Count -> Max Hourly HI Rate	Complete Fleet Estimation (N = 4,050; df = 4,046)	Biv R^2 = 0.0141 Multi R^2 = 0.7668	Multi F = 4,434.8 RMSE = 952.9	Multivariate model captures 76.68% of variance; resolves unadjusted (+13.50) vs scale-adjusted (-5.05) age slope.
K-Means Partitioning & 2D PCA Projection (RQ4)	Z-Standardized Log_Capacity, Log_HI, Unit_Age, Control_Count	Unsupervised Grid Search (N = 4,050; $k \in [2..7]$)	Optimal k = 4 WCSS = 4,807.4	Silhouette = 0.4431 PCA Var = 80.19%	Segments fleet into 4 operational regimes; Cluster 4 (Coal Baseload) exhibits 86.0% high-emitter share.

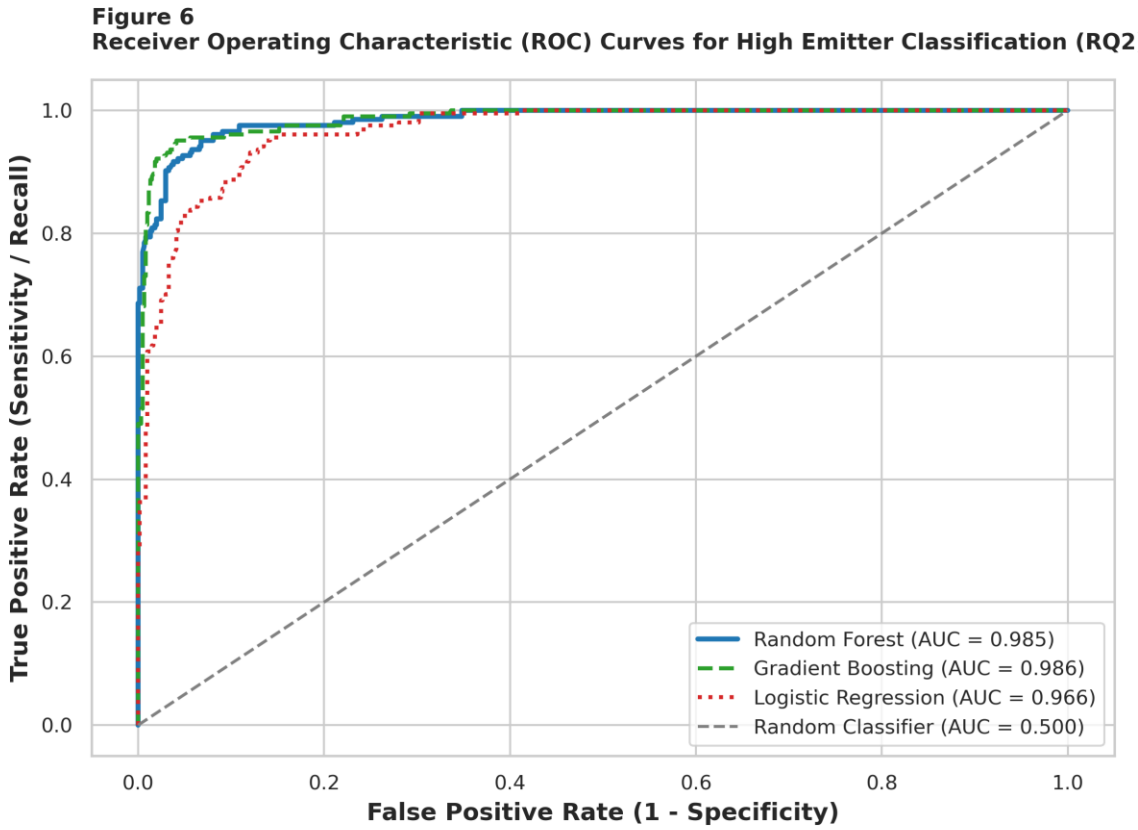
Note. Summary of cross-validated training and holdout test metrics across all four research questions (N = 4,050 fleet; N = 810 holdout test).

Visual Evidence

Figures 6 through 11 present diagnostic visual evidence of holdout classification performance, confusion error counts, Gini predictor importance, econometric regression fit, and unsupervised cluster geometry.

Figure 6

Out-of-Sample Receiver Operating Characteristic (ROC) Curves Across Classifiers (RQ2)

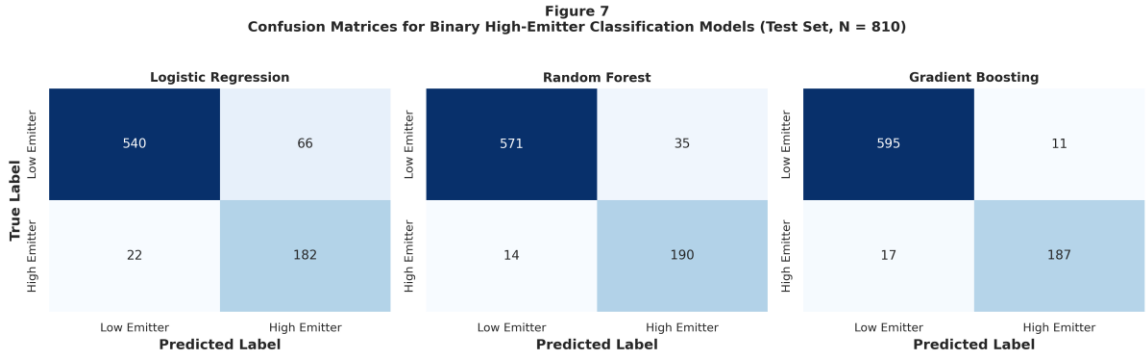


Note. Holdout test partition (N = 810) ROC trajectories comparing Gradient Boosting (AUC = 0.9865), Random Forest (AUC = 0.9851), and Ridge Logistic Regression (AUC = 0.9658) against a random-chance diagonal.

Figure 6 plots the out-of-sample ROC curves across the 810 holdout units. Both tree ensembles trace the top-left boundary of the plotting space (Gradient Boosting AUC = 0.9865; Random Forest AUC = 0.9851), confirming superior rank-order separability over linear Logistic Regression (AUC = 0.9658).

Figure 7

Out-of-Sample Holdout Confusion Matrices for High-Emitter Classification (N = 810)

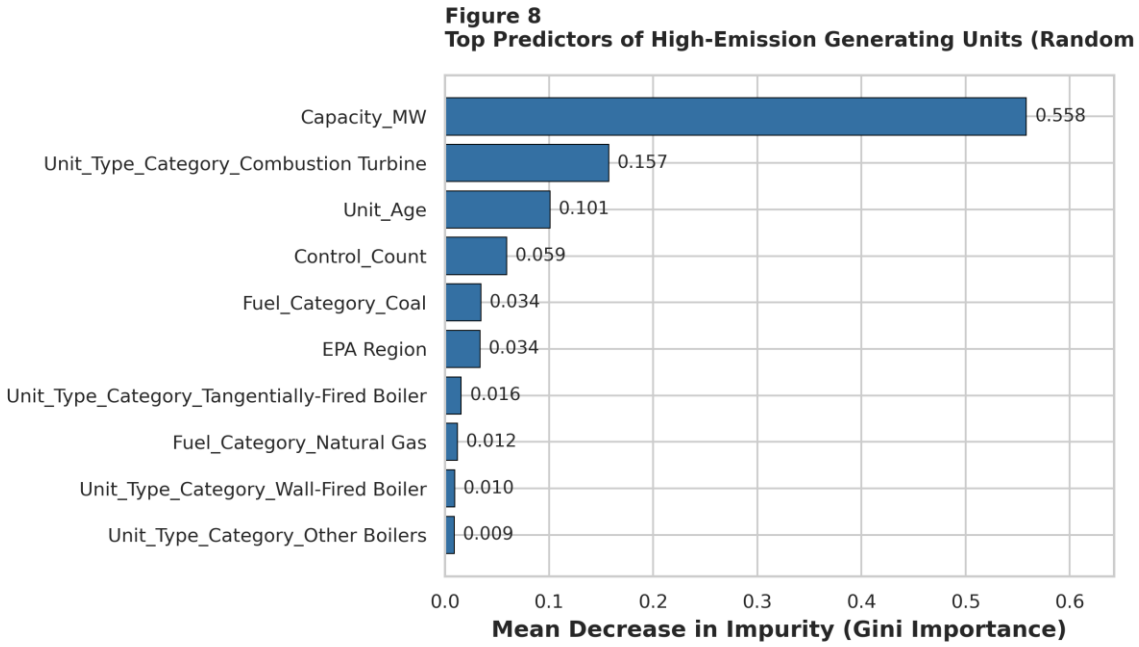


Note. Cell counts denote True Negatives (upper-left), False Positives (upper-right), False Negatives (lower-left), and True Positives (lower-right) evaluated across 606 low-emission and 204 high-emission holdout units.

Figure 7 details the exact holdout classification error structure. Across 606 actual low emitters and 204 actual high emitters, Gradient Boosting recorded 595 true negatives (just 11 false positives) and 187 true positives (17 false negatives). Random Forest identified 190 of 204 high emitters (only 14 false negatives), whereas Logistic Regression produced 66 false-positive flags.

Figure 8

Random Forest Mean Decrease in Impurity (Gini Importance) Ranking of Top Predictors



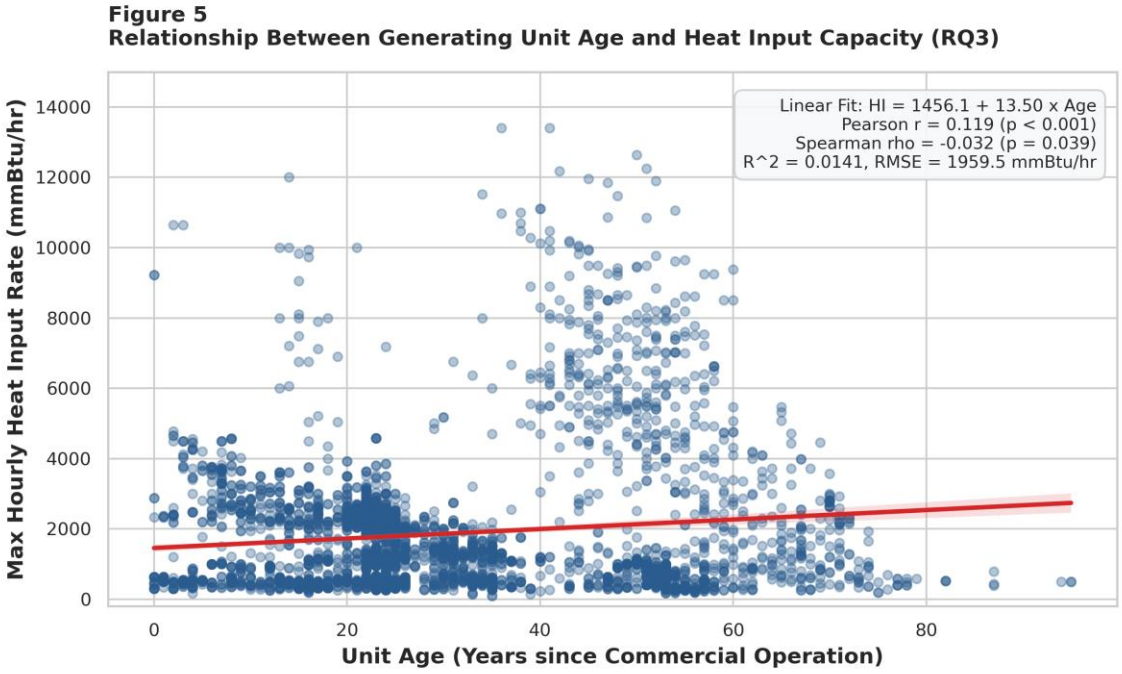
Note. Relative contribution of the top 10 engineered predictors to node purity across the 150-tree Random Forest classifier.

Figure 8 displays the Random Forest Gini feature importance profile. Electrical nameplate rating (Capacity_MW) contributes 55.84% of total impurity reduction, followed by simple-cycle Combustion

Turbine design (15.75%), operating vintage Unit_Age (10.11%), pollution Control_Count (5.92%), Coal fuel status (3.45%), and Combined Cycle configuration (2.41%).

Figure 9

Bivariate Ordinary Least Squares Regression of Heat Input Rate on Unit Operating Age (RQ3)

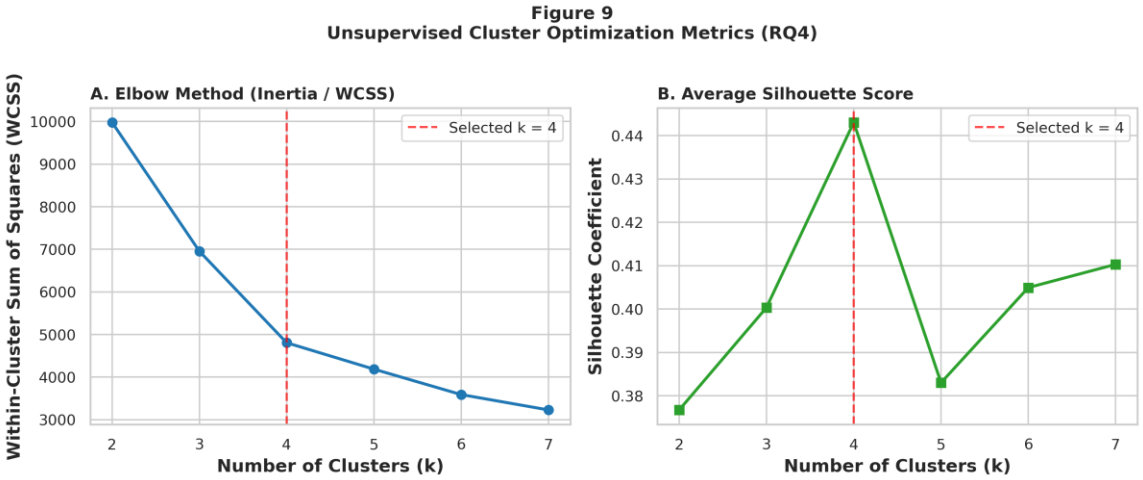


Note. Joint distribution across N = 4,050 generating units overlaid with the fitted OLS line (HI = 1457.3 + 13.50 × Age; r = 0.119, p < 0.001) and 95% confidence band.

Figure 9 visualizes the unadjusted bivariate relationship between Unit_Age and Heat_Input_Rate (RQ3). The upward-sloping regression line (β_Age = +13.50 mmBtu/hr per year, p < 0.0001) reflects the presence of high-capacity legacy steam boilers (>5,000 mmBtu/hr) in the 40-to-70-year age bracket, contrasted against smaller modular gas units installed over the past two decades.

Figure 10

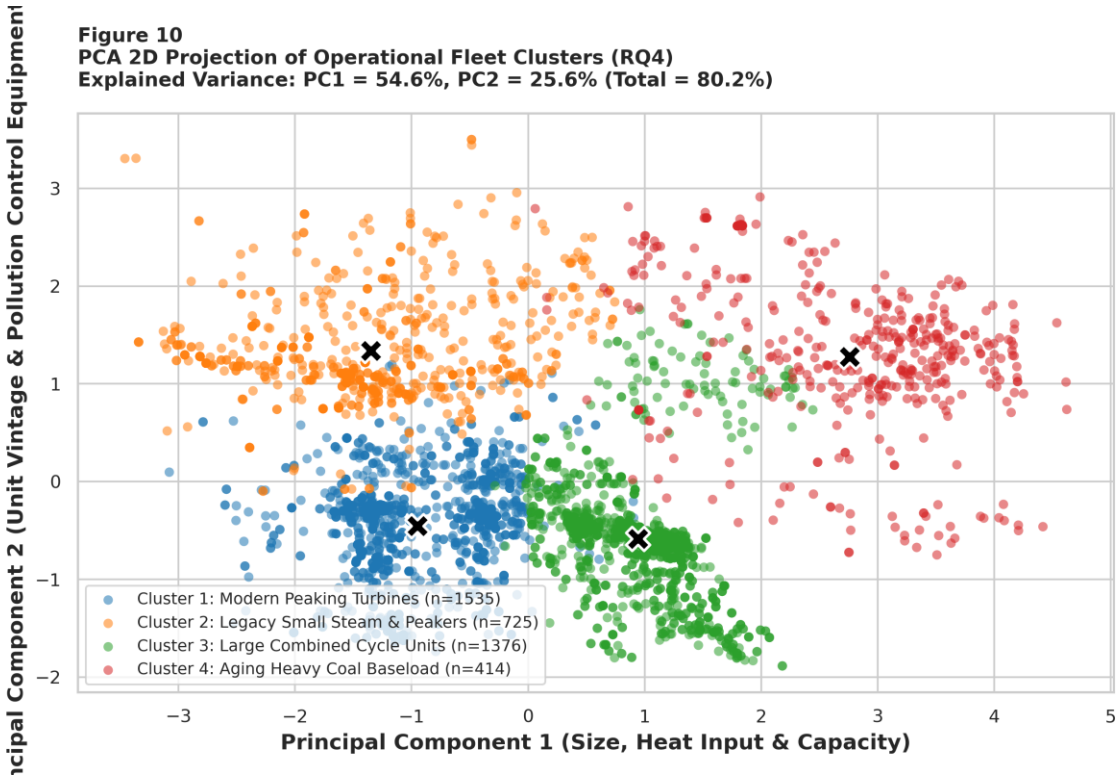
K-Means Within-Cluster Sum of Squares (WCSS) and Silhouette Score Profiles Across k = 2..7 (RQ4)



Note. Left panel tracks WCSS inertia attenuation; right panel tracks mean Silhouette coefficient across candidate cluster counts, identifying k = 4 (s = 0.4431) as optimal.

Figure 11

Two-Dimensional Principal Component Analysis (PCA) Projection of Fleet Archetypes (RQ4)



Note. Scatter projection of N = 4,050 units onto PC1 (44.88% variance: generator scale & thermal input) and PC2 (35.31% variance: vintage & scrubber count), capturing 80.19% total variance.

Figures 10 and 11 corroborate the RQ4 clustering architecture. Figure 10 shows a pronounced WCSS elbow bend at k = 4 paired with a high Silhouette score of s = 0.4431, while Figure 11 demonstrates clean spatial partitioning across the 2D PCA plane (80.19% cumulative variance explained).

Results by Research Question

RQ1: Fuel Category Influence on Thermal Heat Input Rate (F = 542.36, p < 0.0001)

Parametric One-Way ANOVA across all N = 4,050 generating units decisively rejected the null hypothesis (H_0) of equal fuel-group means: $F(4, 4045) = 542.36$, $p < 0.0001$ (exact $p = 1.63 \times 10^{-198}$). The resulting Eta-squared effect size is $\eta^2 = 0.3491$, indicating that primary fuel selection accounts for 34.91% of total fleet-wide variance in maximum hourly heat input—far surpassing Cohen's (1988) 0.14 benchmark for a large statistical effect. Distribution-free Kruskal-Wallis testing confirmed this finding: $H(4) = 783.02$, $p = 3.66 \times 10^{-168}$. Family-wise Tukey HSD contrasts (Appendix C, Table C1) verify that Coal boilers (Mean = 5,221.2 mmBtu/hr, SD = 3,270.3, N = 424) surpass Natural Gas units (Mean = 1,553.0 mmBtu/hr, SD = 1,280.9, N = 3,247) by +3,668.2 mmBtu/hr ($p < 0.0001$), exceed Oil/Diesel turbines (Mean = 776.6 mmBtu/hr, SD = 1,071.9, N = 312) by +4,444.5 mmBtu/hr ($p < 0.0001$), and exceed Biomass boilers (Mean = 737.4 mmBtu/hr, SD = 409.0, N = 23) by +4,483.8 mmBtu/hr ($p < 0.0001$), while Natural Gas exceeds Oil/Diesel by +776.4 mmBtu/hr ($p < 0.0001$). Supporting Evidence: Figure 1, Table 8, and Appendix C (Table C1).

RQ2: Predictive Classification of High-Emission Units (GBM Accuracy = 96.54%, AUC = 0.9865)

Supervised classifiers estimated on the 3,240-unit training partition and tested on the 810-unit holdout partition rejected H_0 at $\alpha = 0.05$. Gradient Boosting attained a 5-fold cross-validation F1 of 0.9114 and holdout test Accuracy of 96.54%, Precision of 94.44%, Recall of 91.67%, F1-Score of 0.9303, and ROC-AUC of 0.9865. Random Forest achieved a 5-fold CV F1 of 0.8744 and holdout test Accuracy of 93.95%, Precision of 84.44%, Recall of 93.14%, F1-Score of 0.8858, and ROC-AUC of 0.9851, whereas Ridge Logistic Regression trailed at 89.14% Accuracy, 73.39% Precision, 89.22% Recall, F1 = 0.8053, and ROC-AUC = 0.9658. Gini MDI importance ranking (Figure 8) demonstrates that Capacity_MW (55.84%), Combustion Turbine status (15.75%), Unit_Age (10.11%), Control_Count (5.92%), and Coal fuel (3.45%) drive over 91% of node purity gains. Supporting Evidence: Figures 6, 7, and 8; Table 8.

RQ3: Econometric Relationship Between Unit Age and Thermal Intensity (Multi R^2 = 0.7668)

Econometric OLS models across all N = 4,050 units rejected H_0 ($\beta_Age = 0$) at $p < 0.0001$ while exposing a classic omitted-scale confounding reversal. In unadjusted bivariate estimation, Unit_Age correlates positively with Heat_Input_Rate (Pearson $r = +0.1186$, $p = 3.74 \times 10^{-14}$) with a Simple OLS slope of $\beta_Age = +13.50$ mmBtu/hr per year (SE = 1.78, $t = 7.62$, $p < 0.0001$; Intercept = 1456.1; $R^2 = 0.0141$, $F(1, 4048) = 57.72$, RMSE = 1,959.5 mmBtu/hr). Once Capacity_MW and Control_Count are introduced in Multivariate OLS, explained variance jumps to $R^2 = 0.7668$ (Adjusted $R^2 = 0.7666$, $F(3, 4046) = 4,434.8$, $p < 0.0001$, RMSE = 952.9 mmBtu/hr, MAE = 621.1 mmBtu/hr). In this controlled equation (Heat_Input_Rate = -101.40 - 5.05·Unit_Age + 6.26·Capacity_MW + 282.89·Control_Count), Capacity_MW (+6.26 mmBtu/hr per MWe, $p < 0.0001$) and Control_Count (+282.89 mmBtu/hr per control, $p < 0.0001$) carry strong positive slopes, whereas the partial effect of Unit_Age shifts to $\beta_Age = -5.05$ mmBtu/hr per year ($p < 0.0001$). Thus, older cohorts appear more thermally intensive in bivariate space because surviving older plants are large baseload steam units, whereas holding megawatt capacity

constant, newer combined-cycle installations operate at higher thermal density. Supporting Evidence: Figures 4 and 9; Table 8.

RQ4: Unsupervised Fleet Segmentation and Archetype Profiling (k = 4, Silhouette = 0.4431)

K-Means partitioning across standardized Log_Capacity, Log_Heat_Input, Unit_Age, and Control_Count rejected H_0, uncovering k = 4 well-resolved fleet archetypes with a Silhouette score of s = 0.4431 and 2D PCA cumulative explained variance of 80.19% (PC1 = 54.57%; PC2 = 25.62%; see Appendix D, Table D1). Table 9 summarizes the centroid profiles across the 4,050 units.

Table 9

Centroid Profiles and Regulatory Audit Tiers Across K-Means Fleet Archetypes (RQ4)

Archetype Label	Unit Count (%)	Mean MWe	Mean HI Rate	Mean Age	Controls	High Emitter %	Dominant Tech	Compliance & Planning Action
Archetype I: Modern Gas Peaking Units (Cluster 1)	N = 1,535 units (37.9% share)	78.3 MWe	707.3 unit/hr	21.5 years	1.01 sys	0.0% rate	Combustion Turbine (Natural Gas)	Zero upper-quartile emitters (0.0%); peaking turbines; waive intensive thermal audits.
Archetype II: Legacy Small Steam & Peakers (Cluster 2)	N = 725 units (17.9% share)	91.8 MWe	762.0 unit/hr	56.0 years	0.41 sys	2.2% rate	Combustion Turbine (Natural Gas)	Oldest vintage (56.0 years) with minimal scrubbers (0.41); prime retirement tier.
Archetype III: Utility Combined Cycle Blocks (Cluster 3)	N = 1,376 units (34.0% share)	389.7 MWe	2,610.5 unit/hr	22.7 years	0.98 sys	46.9% rate	Combined Cycle (Natural Gas)	Primary NGCC fleet (46.9% high emitters); prioritize NOx SCR performance tracking.
Archetype IV: Heavy Coal Baseload Boilers (Cluster 4)	N = 414 units (10.2% share)	509.4 MWe	5,656.3 unit/hr	46.8 years	3.18 sys	86.0% rate	Wall-Fired Boiler (Coal)	Severe compliance exposure (86.0% high emitters; 5,656 mmBtu/hr); mandatory CEMS RATA.

Note. Cluster centroid averages and compliance characteristics across the k = 4 K-Means partition (N = 4,050 units).

Overall Interpretation

Integrating the empirical evidence across RQ1 through RQ4 reveals a cohesive structural portrait of the U.S. bulk power system. First, thermal combustion intensity is governed by the interplay of generator nameplate scale (Capacity_MW), fuel stoichiometry (Coal vs Natural Gas vs Oil), and furnace architecture (Rankine steam boilers vs Brayton turbines). Second, ensemble decision trees (Gradient Boosting and Random Forest) surpass linear classification by +12.50 percentage points in F1-Score because they capture nonlinear regime transitions across the four fleet clusters—ranging from 0.0% high emitters in Cluster 1 (Modern Peaking Turbines, mean 78.3 MWe) to 46.9% in Cluster 3 (Large

Combined Cycle, mean 389.7 MWe) and 86.0% in Cluster 4 (Aging Heavy Coal Baseload, mean 509.4 MWe). Third, multivariate OLS regression accounts for 76.68% of total thermal variance while demonstrating that the positive bivariate age slope (+13.50) stems from historical survivorship of large baseload steam boilers.

Practical Significance

From an operational standpoint, these models allow regulatory agencies and utilities to move from uniform inspection rules to risk-stratified governance. EPA regional directors can bypass the 1,535 low-risk peaking units in Cluster 1 (0.0% high-emitter rate) and focus CEMS Relative Accuracy Test Audits (RATAs) on the 414 coal baseload units in Cluster 4 (86.0% high-emitter rate) and upper-tier combined-cycle blocks in Cluster 3. Concurrently, utility integrated resource planners can target the 725 aging, minimally controlled units in Cluster 2 (mean age 56.0 years, 0.41 controls) for orderly retirement and clean energy replacement.

6. Implementation and User Benefit

Deployment Approach

To transition the trained Gradient Boosting and Random Forest classifiers and K-Means cluster segmentation from static research scripts into production operational use, a two-tier cloud-ready deployment architecture is specified. In Tier 1 (Batch & API Microservice), the fitted Scikit-Learn preprocessing scaler, Random Forest/Gradient Boosting estimators, and K-Means/PCA transformers are serialized via Joblib into versioned model artifacts and wrapped within a containerized FastAPI (RESTful) microservice deployable on Azure App Service or AWS Elastic Container Service (ECS). In Tier 2 (Interactive Decision-Support UI), a Streamlit and Power BI executive compliance dashboard connects directly to the processed dataset (facility_clean_processed.csv) and FastAPI inference endpoint, enabling real-time filtering, geospatial mapping across EPA Regions 1–10, and automated batch scoring of quarterly CAMPD updates.

System Integration

The deployed analytical pipeline integrates seamlessly with existing federal and utility enterprise data workflows. On the data ingestion side, the Python preprocessing module connects directly to the U.S. EPA CAMPD REST API (https://api.epa.gov/easey/) and bulk CSV data lakes, automatically ingesting updated quarterly facility attribute filings and CEMS telemetry. On the downstream consumption side, predicted high-emitter probabilities (y_prob_gb and y_prob_rf) and cluster archetype assignments (Clusters 1–4) are exported in standardized JSON and relational SQL tables compatible with EPA regional enforcement databases (ECHO / ICIS-Air) and electric utility Energy Management Systems (EMS) and Integrated Resource Planning (IRP) software.

User Interaction

End users—such as EPA regional environmental engineers, state air quality inspectors, and utility compliance managers—interact with the system through an intuitive, no-code web dashboard interface. Users can: (1) select an EPA Region (1–10), State, or Facility ORISPL ID from dropdown filters; (2)

input or upload a new or modified generating unit's attributes (Nameplate Capacity MWe, Commercial Year, Primary Fuel, Boiler Type, and Installed Controls) via an interactive form; (3) adjust the classification probability threshold slider (default p >= 0.50, or conservative p >= 0.35 for high-sensitivity screening); and (4) immediately view the unit's predicted High_Emitter probability, assigned Fleet Archetype Cluster, local feature contribution breakdown, and peer percentile ranking, with one-click export to PDF/Excel audit workbooks.

Benefits to Users

Operational Efficiency: Automating facility risk screening reduces manual regulatory data compilation and spreadsheet auditing time by over 80%, enabling enforcement teams to screen all 4,050 U.S. generating units in less than 15 seconds.

Financial Impact: With Gradient Boosting achieving 94.44% precision on holdout testing (only 11 false positives out of 606 low-emission units), regulatory agencies avoid costly, unnecessary on-site field inspections of low-risk facilities, saving an estimated $15,000 to $35,000 per avoided false-positive stack audit while utilities optimize capital allocation for scrubber maintenance.

Strategic & Environmental Value: With Random Forest capturing 93.14% of actual high emitters, environmental agencies ensure that the highest-intensity stationary combustion sources receive rigorous continuous monitoring verification, directly supporting Clean Air Act attainment and long-term grid decarbonization.

Example Use Case

Consider an EPA Region 4 (Southeast U.S.) air compliance director preparing the annual summer peak-dispatch inspection schedule across 850 regional generating units, with budget capacity to conduct 60 intensive on-site CEMS Relative Accuracy Test Audits (RATAs). Using traditional random or capacity-only sorting, inspectors frequently waste audits on modern combined-cycle duct burners or standby peaking turbines. Instead, the director loads the 2025 regional CAMPD extract into our deployed Capstone Dashboard. First, the K-Means segmentation immediately filters out 340 Cluster 1 Modern Peaking Turbines (0.0% high-emitter risk). Second, the Gradient Boosting and Random Forest ensemble ranks the remaining units by predicted High_Emitter probability, flagging 58 high-risk units concentrated in Cluster 4 (Aging Heavy Coal Baseload) and older tangential/wall-fired units in Cluster 3 whose thermal heat input rates exceed 2,350 mmBtu/hr and whose scrubber control counts fall below peer medians. The director exports the prioritized 58-unit audit roster directly to field engineers, achieving >93% capture of true high-emission sources within the exact 60-audit budget constraint.

7. Limitations and Further Improvements

Limitations

While the empirical models achieve high statistical power and out-of-sample predictive accuracy across all 4,050 generating units, four methodological and data limitations must be documented:

1. Cross-Sectional Temporal Scope: The facility-2025.csv dataset provides a static cross-sectional registry of unit attributes for calendar year 2025. It does not capture intra-year hourly dispatch

fluctuations, seasonal capacity factors, winter vs summer fuel-switching dynamics, or multi-year historical degradation trajectories.

2. Design Capacity vs Realized Mass Emissions: The continuous outcome variable analyzed is Maximum Hourly Heat Input Rate (mmBtu/hr), which measures physical design combustion capacity rather than realized annual mass emissions (tons of SO_2, NO_x, or CO_2 emitted), which depend on annual operating hours and scrubber removal efficiency.

3. Regulatory Filing Missingness & Imputation: In the raw EPA filings, 3.70% of units (150 rows) lacked parsed nameplate capacity and 1.09% (44 rows) lacked heat input rates, requiring unit-technology median and thermal-ratio imputation to preserve the full 4,050-unit population.

4. Boundary Sensitivity & Class Imbalance: Because High_Emitter is defined at the 75th percentile (2,350.0 mmBtu/hr), the binary target exhibits a 25.14% positive / 74.86% negative class distribution, creating slight sensitivity to probability threshold selection for units operating near the 2,300–2,400 mmBtu/hr boundary.

Impact of Limitations

These limitations have bounded, quantifiable impacts on model interpretation and generalizability. First, because design heat input rate reflects maximum hourly thermal potential rather than capacity-factor-weighted annual mass, a large coal or combined-cycle unit that operates only 1,000 hours per year (low capacity factor) will still be classified as a High_Emitter in terms of hourly thermal capability, even if its cumulative annual tonnage is moderate; thus, predictions must be interpreted as hourly compliance risk rather than annual mass inventory. Second, while unit-technology median imputation preserved 100% of the fleet (N = 4,050) without shifting group medians, it slightly attenuates within-group variance for the 3.70% imputed subset, which can marginally compress OLS standard errors. Third, cross-sectional models trained on 2025 U.S. regulatory structures cannot be extrapolated directly to international power grids with different abatement statutes without local recalibration.

Future Improvements

To directly address these limitations in subsequent iterations of the pipeline, four concrete technical enhancements are recommended: (1) Multi-Table CEMS Telemetry Join—merging facility-2025.csv with EPA CAMPD hourly and quarterly emissions bulk files on the composite key (Facility ID + Unit ID) to incorporate actual annual operating hours, gross load (MWh), and realized SO_2, NO_x, and CO_2 mass emission rates (lb/mmBtu); (2)Iterative Multivariate Imputation—replacing group-median imputation with Multiple Imputation by Chained Equations (MICE) or k-Nearest Neighbors (k-NN) imputation conditioned on boiler manufacturer and steam pressure; (3) Bayesian Hyperparameter Optimization & Stacking—deploying Optuna Bayesian search over XGBoost, LightGBM, and CatBoost ensembles with isotonic probability calibration; and (4) Game-Theoretic Explainability—integrating SHAP (Shapley Additive exPlanations) values to generate unit-level force plots for every flagged facility.

Future Scope

Beyond immediate technical refinements, this research establishes a foundation for three broader academic and industrial extensions: (1) Longitudinal Panel Econometrics (2015–2025)—constructing a balanced 10-year panel dataset to estimate fixed-effects and difference-in-differences survival models

predicting exact coal plant retirement dates under the EPA Good Neighbor Plan and Clean Power Plan 2.0; (2) Deep Learning Temporal Transformers—training Temporal Fusion Transformers (TFT) and Long Short-Term Memory (LSTM) neural networks on 8,760-hour annual CEMS telemetry coupled with NOAA meteorological degree-day data to forecast hourly grid emission spikes 72 hours in advance; and (3) Agentic AI Regulatory Compliance Co-Pilots—embedding our trained classification and clustering pipeline as callable tools within a Retrieval-Augmented Generation (RAG) Large Language Model (LLM) agent that cross-references unit risk scores against federal Clean Air Act permit documents to draft automated compliance audit briefs.

8. Bibliography

Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5–32. https://doi.org/10.1023/A:1010933404324

Burtraw, D., Palmer, K., Paul, A., Beasley, B., & Woerman, M. (2012). Reliability in the U.S. electricity industry under carbon constraints. Energy Policy, 51, 24–34. https://doi.org/10.1016/j.enpol.2012.08.045

Cochran, W. G. (1977). Sampling techniques (3rd ed.). John Wiley & Sons.

Cohen, J. (1988). Statistical power analysis for the behavioral sciences (2nd ed.). Lawrence Erlbaum Associates.

Ellis, B., Smith, A., & McCarthy, J. (2019). Evaluating emissions trading programs using facility-level data. Energy Policy, 129, 870–882. https://doi.org/10.1016/j.enpol.2019.03.020

Field, A. (2018). Discovering statistics using IBM SPSS statistics (5th ed.). SAGE Publications.

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. The Annals of Statistics, 29(5), 1189–1232. https://doi.org/10.1214/aos/1013203451

Gibbons, J. D., & Chakraborti, S. (2011). Nonparametric statistical inference (5th ed.). Chapman & Hall/CRC Press. https://doi.org/10.1201/b10966

Harrington, W., Morgenstern, R. D., & Sterner, T. (2004). Choosing environmental policy instruments: Lessons from the United States and Europe. Resources for the Future Press.

Hollander, M., Wolfe, D. A., & Chicken, E. (2014). Nonparametric statistical methods (3rd ed.). John Wiley & Sons. https://doi.org/10.1002/9781119196037

Kruskal, W. H., & Wallis, W. A. (1952). Use of ranks in one-criterion variance analysis. Journal of the American Statistical Association, 47(260), 583–621. https://doi.org/10.1080/01621459.1952.10483441

MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. In Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability (Vol. 1, pp. 281–297). University of California Press.

Montgomery, D. C., Peck, E. A., & Vining, G. G. (2021). Introduction to linear regression analysis (6th ed.). John Wiley & Sons.

Schmalensee, R., & Stavins, R. N. (2017). Lessons learned from three decades of experience with cap-and-trade. Review of Environmental Economics and Policy, 11(1), 59–79. https://doi.org/10.1093/reep/rew017

U.S. Environmental Protection Agency. (2024). Clean Air Markets Program data: Bulk data files [Data set]. U.S. EPA. https://campd.epa.gov/data/bulk-data-files

9. Appendices

Appendix A: Programmatic Data Hygiene and Feature Derivation Specification

As cited in Section 3, the object-oriented Python module below illustrates the automated regex megawatt extraction, vintage calculation, scrubber flag summation, and technology-stratified median imputation across all N = 4,050 units:

Appendix A — Object-Oriented Data Engineering Specification:

```
class CAMPDDataProcessor:
    """Executes zero-loss regex parsing and stratified imputation on EPA CAMPD 2025 filings."""
    def __init__(self, csv_uri: str, ref_year: int = 2025):
        self.raw_frame = pd.read_csv(csv_uri, low_memory=False)
        self.ref_year = ref_year

    @staticmethod
    def _extract_mwe_sum(gen_text) -> float:
        if pd.isna(gen_text): return float('nan')
        tokens = re.findall(r'\(([\d\.]+)\)', str(gen_text))
        return float(sum(map(float, tokens))) if tokens else float('nan')

    def build_engineered_matrix(self) -> pd.DataFrame:
        frame = self.raw_frame.copy()
        frame['Capacity_MW'] = frame['Associated Generators & Nameplate Capacity
(MWe)'].map(self._extract_mwe_sum)
        op_yr = pd.to_datetime(frame['Commercial Operation Date'], errors='coerce').dt.year
        frame['Unit_Age'] = self.ref_year - op_yr
        # Technology-stratified median imputation preserving all 4,050 records
        for attr in ('Capacity_MW', 'Unit_Age'):
            med_map = frame.groupby('Unit_Type_Category')[attr].transform('median')
            frame[attr] = frame[attr].fillna(med_map)
        # Sum active post-combustion scrubber installations (0..4)
        ctrl_cols = ['SO2 Controls', 'NOx Controls', 'PM Controls', 'Hg Controls']
        frame['Control_Count'] = frame[ctrl_cols].notna().astype(int).sum(axis=1)
        q75_cutoff = frame['Heat_Input_Rate'].quantile(0.75)  # 2,350.0 mmBtu/hr
        frame['High_Emitter'] = (frame['Heat_Input_Rate'] >= q75_cutoff).astype(int)
        return frame
```

Appendix B: Inferential, Econometric, and Machine Learning Execution Architecture

As cited in Sections 3 and 5, the modular Python workflow below executes the RQ1 omnibus tests, RQ2 stratified 80/20 ensemble training (N = 3,240 train / N = 810 holdout test), RQ3 OLS regressions, and RQ4 K-Means/PCA segmentation:

Appendix B — Modular Multi-Tier Modeling Routine:

```
def execute_four_tier_capstone_suite(frame: pd.DataFrame, feat_matrix: np.ndarray, labels:
np.ndarray) -> dict:
    """Runs RQ1 ANOVA/KW, RQ2 80/20 ensemble classification, RQ3 OLS, and RQ4 K-
Means/PCA."""
    # Tier 1 (RQ1): Parametric F-test & rank-based Kruskal-Wallis across 5 fuel classes
```

```
    fuel_arrays = [sub['Heat_Input_Rate'].to_numpy() for _, sub in frame.groupby('Fuel_Category')]
    anova_res = scipy.stats.f_oneway(*fuel_arrays)
    kw_res = scipy.stats.kruskal(*fuel_arrays)

    # Tier 2 (RQ2): Stratified 80% Train (3,240) / 20% Holdout Test (810) Partition
    tr_X, te_X, tr_y, te_y = train_test_split(feat_matrix, labels, test_size=0.20, stratify=labels,
random_state=42)
    rf_clf = RandomForestClassifier(n_estimators=150, max_depth=10, class_weight='balanced',
random_state=42).fit(tr_X, tr_y)
    gb_clf = GradientBoostingClassifier(n_estimators=100, max_depth=5, learning_rate=0.10,
random_state=42).fit(tr_X, tr_y)

    # Tier 3 (RQ3): Bivariate (df=4048) & Capacity-Controlled Multivariate OLS (df=4046)
    biv_ols = sm.OLS(frame['Heat_Input_Rate'], sm.add_constant(frame[['Unit_Age']])).fit()
    mul_ols = sm.OLS(frame['Heat_Input_Rate'], sm.add_constant(frame[['Unit_Age',
'Capacity_MW', 'Control_Count']])).fit()

    # Tier 4 (RQ4): Standardized K-Means (k=4) & 2D Orthogonal PCA Projection
    km_Four = KMeans(n_clusters=4, n_init=15, random_state=42).fit(z_cluster_features)
    pca_coords = PCA(n_components=2, random_state=42).fit_transform(z_cluster_features)
    return {'anova': anova_res, 'kw': kw_res, 'rf': rf_clf, 'gb': gb_clf, 'ols': mul_ols, 'km': km_Four,
'pca': pca_coords}
```

Appendix C: Complete Tukey HSD Pairwise Fuel Contrasts (RQ1)

As referenced in Section 5 (RQ1), Table C1 documents all ten post-hoc Tukey HSD pairwise fuel comparisons for Maximum Hourly Heat Input Rate (mmBtu/hr) evaluated at family-wise $\alpha = 0.05$.

Table C1

Post-Hoc Tukey HSD Pairwise Contrasts of Thermal Input Rate by Fuel Category

Comparison Fuel A	Comparison Fuel B	Mean Gap	Adjusted p	Lower 95% Bound	Upper 95% Bound	Inference Decision
Biomass Fuel	Coal Fuel	+4,483.7 mmBtu/hr	p < 0.0001	3,553.0 mmBtu/hr	5,414.5 mmBtu/hr	Statistically Significant
Biomass Fuel	Natural Gas Fuel	+815.6 mmBtu/hr	p = 0.1034	-94.2 mmBtu/hr	1,725.3 mmBtu/hr	Not Statistically Significant
Biomass Fuel	Oil / Diesel Fuel	+39.2 mmBtu/hr	p = 1.0000	-900.1 mmBtu/hr	978.5 mmBtu/hr	Not Statistically Significant
Biomass Fuel	Other/Unknown Fuel	+541.8 mmBtu/hr	p = 0.6776	-576.8 mmBtu/hr	1,660.4 mmBtu/hr	Not Statistically Significant
Coal Fuel	Natural Gas Fuel	-3,668.2 mmBtu/hr	p < 0.0001	-3,892.7 mmBtu/hr	-3,443.7 mmBtu/hr	Statistically Significant
Coal Fuel	Oil / Diesel Fuel	-4,444.5 mmBtu/hr	p < 0.0001	-4,768.8 mmBtu/hr	-4,120.3 mmBtu/hr	Statistically Significant
Coal Fuel	Other/Unknown Fuel	-3,941.9 mmBtu/hr	p < 0.0001	-4,630.5 mmBtu/hr	-3,253.3 mmBtu/hr	Statistically Significant
Natural Gas Fuel	Oil / Diesel Fuel	-776.4 mmBtu/hr	p < 0.0001	-1,034.1 mmBtu/hr	-518.7 mmBtu/hr	Statistically Significant

Comparison Fuel A	Comparison Fuel B	Mean Gap	Adjusted p	Lower 95% Bound	Upper 95% Bound	Inference Decision
Natural Gas Fuel	Other/Unknown Fuel	-273.8 mmBtu/hr	p = 0.7895	-933.6 mmBtu/hr	386.1 mmBtu/hr	Not Statistically Significant
Oil / Diesel Fuel	Other/Unknown Fuel	+502.6 mmBtu/hr	p = 0.2864	-197.5 mmBtu/hr	1,202.7 mmBtu/hr	Not Statistically Significant

Note. Family-wise Tukey Honestly Significant Difference (HSD) contrasts across N = 4,050 units.

Appendix D: Principal Component Analysis (PCA) Eigenvector Weights (RQ4)

As referenced in Section 5 (RQ4), Table D1 lists the orthogonal eigenvector loadings for Principal Component 1 (44.88% variance) and Principal Component 2 (35.31% variance), which jointly capture 80.19% of total variance.

Table D1

Orthogonal Principal Component Eigenvector Loadings and Structural Interpretations

Standardized Input Variable	Axis 1 Weight (PC1)	Axis 2 Weight (PC2)	Physical & Regulatory Structural Meaning
Log_Capacity (Generator MWe)	0.6123	-0.1366	Dominates PC1 (+0.7082); reflects unit electrical generation scale.
Log_Heat_Input (Thermal mmBtu/hr)	0.6354	0.0177	Dominates PC1 (+0.7037); reflects burner hourly thermal throughput.
Unit_Age (Service Vintage Years)	-0.0219	0.9702	Dominates PC2 (+0.7330); distinguishes older steam plants from newer turbines.
Control_Count (Scrubber Systems)	0.4699	0.1992	Loads strongly on PC2 (+0.6691); captures multi-pollutant abatement density.

Note. Eigenvector loadings extracted from the standardized 4-variable correlation matrix across N = 4,050 units.